

Gyre: Test-Time Routing for Long-Context Sequence Models

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Abstract

Long-context sequence models assign predictive features to non-uniform internal resources, but pretraining fixes this allocation. We introduce Gyre, a frozen-weight test-time routing framework that rotates signal coordinates at legal internal ports while preserving weights, masks, gates, recurrence rules, and cache coordinates. Gyre derives a Fisher-Gramian alignment objective, proves abstention conditions for flat or already aligned responses, and selects identity, C-FGR, or Stream-FGR under stability and cost constraints. On Qwen3.6-27B, restricted gain predicts held-out NLL reduction, and the selector reaches the deployable quality-efficiency Pareto frontier, improving relative NLL by 0.86% with an 82.2% prompt win rate at 7.0% wall-clock overhead.

1 Introduction

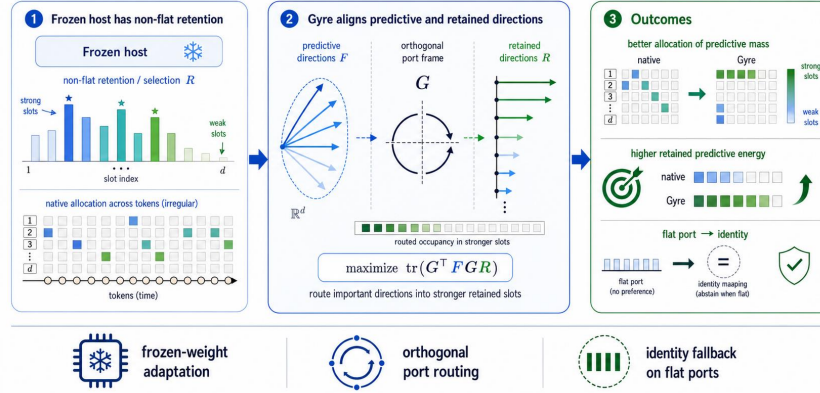


Figure 1: Overview of the Gyre framework.

Hybrid sequence models compress long prompts through finite internal resources: recurrent state channels, gated attention heads, value directions, and output-selection coordinates. These resources are rarely exchangeable. Some directions are retained longer, selected more strongly, or coupled more tightly to downstream logits. Pretraining learns an average assignment between features and these host resources, but a particular prompt can make a different set of directions predictive. We introduce frozen-weight test-time frame routing, a method that rotates signal coordinates at legal internal ports while keeping model weights, gates, recurrence rules, masks, and cache coordinates fixed. The central object is a Fisher-Gramian score, $\text{tr}(G^T FGR)$, which measures how well prompt-local predictive directions align with host retention or selection directions. This gives an admission rule: route only when response is non-flat and restricted Fisher-Gramian gain is positive.

Related work. Gyre is related to work on test-time adaptation, efficient long-context sequence modeling, recurrent and state-space sequence mixers, and deployment-time transformations, but differs in the object being adapted. Test-time training and adaptation methods update parameters, normalization statistics, or hidden-state learning rules at inference time, for example self-supervised test-time training, entropy-minimization adaptation, augmentation-based adaptation, and recent sequence-model TTT variants [28, 33, 40]. Gyre instead keeps the base host fixed and selects an orthogonal signal frame at legal internal ports. Efficient-attention work reduces the cost of long contexts by changing the attention pattern, kernel approximation, memory layout, or cache policy [3–6, 8, 15, 16, 18, 20, 31, 34, 36, 39]. These methods improve how tokens are mixed, stored, or computed; they do not route prompt-local predictive directions into fixed, non-flat host-response directions. A second line designs recurrent, convolutional, retention, or state-space hosts with better long-range memory and decoding cost [1, 7, 9, 12–14, 22, 23, 27, 29, 37]. Gyre is complementary to this line as it assumes such non-uniform hosts already exist and asks whether a frozen host’s retention or selection coordinates are well matched to the current prompt. Finally, PTQ and compression methods use calibration, equivalent transformations, salient-channel protection, or learned quantization parameters to preserve accuracy under low precision [2, 11, 19, 21, 26, 35]. Gyre differs by using rotations to change the allocation of predictive signal to measured host-response directions while respecting fixed gates, masks, recurrence laws, and cache coordinates.

2 Preliminaries

2.1 Frozen Hosts and Internal Ports

We consider an autoregressive sequence model with frozen parameters θ_0 . Given a prefix $x_{\leq t}$, the model produces a distribution over x_{t+1} and incurs next-token loss $\ell_t(\theta_0)$. The word frozen matters here. Gyre does not change θ_0 , does not add trainable adapters during inference, and does not rewrite the recurrence law of the host. We instead look inside the frozen computation graph and ask whether the coordinates through which information enters a selected module are the best coordinates for the current prompt. This question is meaningful only in modern hybrid hosts because their internal channels are rarely exchangeable. A gated attention head, a value-output channel, or a recurrent update coordinate can differ in downstream selection strength, retention time, or sensitivity to gates. Gyre starts from this asymmetry.

A *port* is an internal signal coordinate

$$u_{t,p} \in \mathbb{R}^{d_p},$$

where p indexes a layer and a submodule. We write u_t when the port is clear. The port signal is distinct from the host’s control variables. In a gated attention block, the control variables include attention logits, masks, and gates. In a recurrent block, they include decay, update, routing, and state-transition coefficients. Gyre is allowed to transform the signal coordinate by an orthogonal frame

$$u_t \mapsto G^\top u_t, \quad G \in O(d_p),$$

while keeping the host policy fixed. This distinction is the first closure condition of the framework. If a gauge changes gates, decays, masks, routers, or cache coordinates, then the object that we use to measure host response changes with the gauge itself. In that case the simple Fisher-Gramian objective below becomes circular.

2.2 Predictive Directions

At a port, two positive semidefinite matrices describe the local opportunity for frame routing. The first is the prompt-local predictive Fisher

$$F_p = \mathbb{E}_t[g_t g_t^\top], \quad g_t = \nabla_{u_t} \ell_t.$$

It measures which infinitesimal port directions would change the predictive loss most strongly on the current calibration positions. The second is the host response Gramian

$$R_p = \mathbb{E}_t[J_t^\top J_t],$$

where J_t maps a small perturbation of the port signal to a downstream quantity selected by the host. At a stateless attention-output port, J_t can be the local output map. At a recurrent update port, J_t can be a finite-horizon response map, so R_p measures which input directions survive or influence future states. The key point is that F_p is prompt-local and task-facing, while R_p is host-local and resource-facing. Gyre exists when these two objects are non-flat and misaligned.

72 Given $F, R \succeq 0$, an orthogonal frame G has retained predictive energy

$$\mathcal{J}(G) = \text{tr}(G^\top FGR).$$

73 This surrogate has a simple interpretation. The gauge first maps host response directions into the
 74 coordinate system where the prompt Fisher is measured. The trace then scores how much Fisher
 75 mass lands in high-response host directions. A large value means that directions which matter for
 76 prediction are being written into coordinates that the frozen host tends to preserve, amplify, or select.
 77 A small value means that predictive signal is being routed into weak host directions.

78 **Theorem 1** (Fisher-Gramian alignment). *Let $F, R \succeq 0$ have eigenvalues $\sigma_1 \geq \dots \geq \sigma_d$ and*
 79 *$r_1 \geq \dots \geq r_d$. Then*

$$\max_{G \in O(d)} \text{tr}(G^\top FGR) = \sum_{i=1}^d \sigma_i r_i.$$

80 *A maximizer aligns the eigenvectors of R with the eigenvectors of F in sorted order, up to rotations*
 81 *inside equal-eigenvalue subspaces.*

82 The theorem gives the mathematical content behind the word routing. If $u_{F,i}$ and $u_{R,j}$ are Fisher and
 83 response eigenvectors, then

$$P_{ij}(G) = |u_{F,i}^\top G u_{R,j}|^2$$

84 acts like an assignment matrix between predictive directions and host directions. Orthogonality makes
 85 $P(G)$ doubly stochastic, and the objective becomes

$$\mathcal{J}(G) = \sum_{i,j} \sigma_i r_j P_{ij}(G).$$

86 Thus the best full frame assigns large σ_i to large r_j . This is the first-principles reason Gyre rotates
 87 directions rather than tokens. Tokens already flow through the host. Gyre changes which host
 88 directions receive the prompt’s predictive coordinates.

89 2.3 Abstention and Restricted Gauges

90 The same theorem also tells us when Gyre has no mathematical opportunity. If the host response
 91 is flat: $R = \alpha I$, then $\mathcal{J}(G) = \alpha \text{tr}(F)$ for every orthogonal G . A flat port has no privileged host
 92 directions, so frame routing cannot improve the surrogate. If F and R are already aligned under
 93 identity, a new frame also has little value. This gives an admission rule before any downstream
 94 experiment: Gyre acts only on legal ports with non-flat response and positive predicted restricted
 95 gain.

96 Production gauges rarely range over all of $O(d)$. They may be block permutations, block Givens
 97 rotations, head-block rotations, or quantization-group-aligned rotations. We therefore define the
 98 deployable gain

$$\Delta_{\mathcal{G}} = \max_{G \in \mathcal{G}} \mathcal{J}(G) - \mathcal{J}(I), \quad \mathcal{G} \subset O(d),$$

99 and compare it with the full-frame oracle gain

$$\Delta_{O(d)} = \max_{G \in O(d)} \mathcal{J}(G) - \mathcal{J}(I).$$

100 The realized fraction

$$\eta_{\mathcal{G}} = \frac{\Delta_{\mathcal{G}}}{\Delta_{O(d)} + \varepsilon}$$

101 keeps the claim calibrated. The full alignment theorem explains the direction of the method. The
 102 restricted gain is what a real implementation earns.

103 3 Frozen-Weight Test-Time Frame Routing

104 3.1 From average feature allocation to prompt-local frame choice

105 Section 2 gives the local geometry. At a legal port, the prompt-local Fisher F measures directions
 106 that matter for next-token prediction, while the host response Gramian R measures directions that the
 107 frozen host selects or retains. The Fisher-Gramian score

$$\mathcal{J}(G) = \text{tr}(G^\top FGR)$$

then asks how much predictive mass reaches strong host directions after an orthogonal frame G is applied. The full-frame optimum aligns Fisher eigenvectors with response eigenvectors, and flat-response ports give no routing opportunity. This is the first design principle of Gyre. We adapt only when there is measured geometry to exploit.

3.2 The Frame-routing Intervention

Gyre changes a coordinate frame rather than a weight tensor. Let $u_{t,p} \in \mathbb{R}^{d_p}$ be a legal signal at port p , and let $\mathcal{G}_p \subseteq O(d_p)$ be a production gauge class, such as head-block rotations, block Givens rotations, block permutations, or quantization-group-aligned rotations. Given estimated matrices F_p and R_p , the ideal restricted frame is

$$G_p^* \in \arg \max_{G \in \mathcal{G}_p} \text{tr}(G^\top F_p G R_p).$$

The runtime intervention is $u_{t,p} \mapsto (G_p^*)^\top u_{t,p}$. The base parameters θ_0 remain fixed. The host recurrence, gate policy, attention mask, router, and cache coordinate system remain fixed as well. This separation is what makes the method different from test-time training. TTT-style layers make the hidden state or a subset of weights adapt through a learning rule [10, 30, 32]. Mamba-style work changes the recurrent host architecture itself, for example by improving the state-space recurrence, complex state update, or MIMO structure [17]. Gyre instead adapts the frame through which prompt-predictive directions enter an already trained host.

Theorem 2 (Surrogate admission for frozen frame routing). *Let $F, R \succeq 0$ be fixed matrices for a legal port, and let $\mathcal{G} \subseteq O(d)$ contain the identity. Define*

$$\Delta_{\mathcal{G}} = \max_{G \in \mathcal{G}} \text{tr}(G^\top F G R) - \text{tr}(F R).$$

If $R = \alpha I$, then $\Delta_{\mathcal{G}} = 0$. More generally, a non-identity frame improves the Fisher-Gramian surrogate over identity if and only if there exists $G \in \mathcal{G}$ such that

$$\text{tr}(G^\top F G R) > \text{tr}(F R).$$

The theorem gives the admission rule used by the method. We do not activate Gyre merely because a model exposes a port. A port first needs to be legal, then its response needs to be non-flat, and finally the restricted gauge class needs to produce positive gain with adequate confidence. If these tests fail, identity is the correct action for the local surrogate. This is important for the paper narrative because it makes abstention part of the design. A method that rotates every available tensor would be easier to describe, but it would also erase the distinction between a geometric opportunity and an arbitrary perturbation.

4 Legal Ports and Frame Modes in Hybrid Hosts

4.1 The Legal-port Contract

A frame-routing theorem is only as valid as the port where the frame is placed. We therefore define a legal port by the computation it leaves unchanged. Locally, write the host as

$$H_t(u_t; c_t),$$

where u_t is a signal coordinate and c_t collects control variables such as gates, masks, routing logits, decay parameters, state-transition coefficients, cache addresses, and normalization statistics. A Gyre frame is legal when it applies

$$u_t \mapsto G^\top u_t$$

while leaving c_t fixed. This keeps the response Gramian R tied to a frozen host policy. If a candidate frame changes the gate or recurrence policy that defines R , the objective becomes $\text{tr}(G^\top F G R(G))$, and the Fisher-Gramian alignment theorem no longer applies.

This contract also rules out several tempting shortcuts. A symmetric dense-attention rotation with $Q' = QG$ and $K' = KG$ leaves the logits unchanged because

$$Q' K'^\top = Q G G^\top K^\top = Q K^\top.$$

This is an invariance, not a routing mechanism. A time-varying pre-cache value gauge is also unsafe unless cached entries carry frame metadata or the cache is replayed into a common frame. Otherwise future reads combine vectors stored in different coordinate systems. Gyre therefore focuses on output-side attention ports and recurrent signal-update ports where the port response can be measured without changing the host policy.

4.2 Gated Attention Ports

Gated Attention gives the cleanest stateless host class because it separates a content signal from a learned selection operator. Let

$$o_t = \text{SDPA}(Q, K, V)_t, \quad y_t = W_O D(g_t) o_t,$$

where $D(g_t)$ is a diagonal gate. Recent gated-attention work studies a query-dependent head-specific sigmoid gate after SDPA and reports improvements in scaling, stability, attention-sink behavior, and long-context extrapolation [24]. For Gyre, the gate matters because it makes the channel response non-uniform. The exact response Gramian depends on whether the frame acts before or after the gate.

In *GA-Sel-FGR*, the frame acts on the SDPA output before the gate:

$$\tilde{y}_t = W_O D(g_t) G^\top o_t.$$

The local response from $G^\top o_t$ to output is $W_O D(g_t)$, hence

$$R_{\text{GA}}^{\text{sel}} = \mathbb{E}_t[D(g_t)^\top W_O^\top W_O D(g_t)].$$

This is the gate-selection version of Gyre. It routes prompt-predictive SDPA-output directions into directions that survive the gate and output projection.

In *GA-Out-FGR*, the frame acts after the gate:

$$h_t = D(g_t) o_t, \quad \tilde{y}_t = W_O G^\top h_t.$$

Now the response from $G^\top h_t$ to output is W_O , so

$$R_{\text{GA}}^{\text{out}} = W_O^\top W_O.$$

The gate still changes the distribution and Fisher of h_t , but it is not part of the response Gramian in this coordinate system. We keep GA-Sel-FGR and GA-Out-FGR separate because mixing their Gramians would make the method circular. The first placement studies gate-selected capacity. The second studies output-projection capacity after gating.

4.3 Recurrent Dignal Ports

Recurrent hosts require a stronger coherence condition because their state stores past writes. For a Gated DeltaNet-style port, the legal form is

$$z_{t+1} = A_t z_t + B_t G^\top u_t.$$

The frame acts on the signal or update channel u_t . It does not act on A_t , decay parameters, memory-erasure gates, delta-rule controls, or routing logits. This matches the role of GDN as a host that combines gating for adaptive memory erasure with the delta rule for targeted memory modification [38]. Gyre does not replace those mechanisms. It changes which signal directions are written into the host's existing recurrent coordinates.

This placement also clarifies what would be insufficient evidence. Rotating an arbitrary module output may test output-channel sensitivity, but it does not necessarily test retention allocation. Rotating gate or decay inputs changes the frozen host policy. For recurrent FGR, the port must be the signal path whose perturbations feed the finite-state memory response measured by R . This is the key implementation burden for a Qwen-like hybrid model, whose public model card reports a 64-layer layout with repeated Gated DeltaNet and Gated Attention blocks [25].

4.4 C-FGR

C-FGR uses a fixed frame over a coherent segment:

$$G_t = G_s.$$

For a recurrent host,

$$z_{t+1} = A_t z_t + B_t G_s^\top u_t.$$

Unrolling from the segment boundary gives

$$z_T = \Phi_{0 \rightarrow T} z_0 + \sum_{t < T} \Phi_{t+1 \rightarrow T} B_t G_s^\top u_t.$$

Every write term enters in the same frame. This is the main reason C-FGR is the default recurrent mode: it preserves state-frame coherence. The price is a segment contract. The frame must be known before the segment begins, obtained by fit-replay-freeze, applied only after a prefix boundary with claims restricted to future writes, or used after an explicit state reset. A mid-state switch without replay mixes identity-frame history with newly framed writes.

C-FGR is therefore most natural when the segment is long enough to amortize calibration, validation, and possible replay. It is less attractive for short contexts or rapidly shifting prompts. That tradeoff is not an implementation accident. It follows from the fixed cost of establishing a coherent segment frame.

4.5 Stream-FGR

Stream-FGR lets the frame move gradually:

$$G_{t+1} = G_t \exp(\Omega_t), \quad \Omega_t^\top = -\Omega_t.$$

The skew-symmetric tangent Ω_t keeps the update on the orthogonal group. For stateless GA ports, this motion does not mix old and new recurrent writes because there is no recurrent state at the port. The limiting factors are estimation stability, validation, trust region, fusion, and wall-clock cost.

For recurrent ports, continuous motion creates a state-mixing term. Compare the actual state with a final-frame reference:

$$e_T = \sum_{t < T} \Phi_{t+1 \rightarrow T} B_t (G_t^\top - G_T^\top) u_t.$$

If

$$\|\Phi_{t+1 \rightarrow T}\| \leq K \rho^{T-t-1}, \quad \|B_t\| \leq B, \quad \|u_t\| \leq U, \quad \|G_{s+1} - G_s\| \leq \eta,$$

then a finite-horizon bound scales as

$$\|e_T\| \leq KBU\eta \sum_{h=0}^{H-1} (h+1)\rho^h.$$

For a stable infinite-tail approximation, this becomes

$$\|e_T\| \lesssim \frac{KBU\eta}{(1-\rho)^2}.$$

This bound explains why recurrent Stream-FGR needs certification. The constant K measures transient amplification, and setting $K = 1$ is valid only after a host-specific proof or measurement. Non-normal recurrent dynamics can have large transient growth even with stable eigenvalues, so uncertified recurrent Stream-FGR is rejected by the controller.

4.6 Budgeted Mode Routing

The frame type is chosen by a mode router. For each legal port-band b , the feasible modes are identity I , C-FGR C , and Stream-FGR S . Identity is always feasible. C-FGR is feasible when the segment contract holds. Stream-FGR is feasible at stateless ports under drift and cost checks, and at recurrent ports only under finite-horizon stability certification.

A simple utility model explains the regimes. If a C-FGR frame has per-token surrogate gain Δ_C over a segment of length S , fixed cost C_A , and risk penalty R_C , then

$$B_C(S) = \Delta_C S - C_A - R_C.$$

If Stream-FGR needs time τ to reach its target and accrues gain linearly along the geodesic, then with per-token cost c_S and risk penalty R_S ,

$$B_S(S) = \begin{cases} \frac{\Delta_S S^2}{2\tau} - c_S S - R_S, & S \leq \tau, \\ \Delta_S (S - \tau/2) - c_S S - R_S, & S > \tau. \end{cases}$$

The controller selects

$$m_b^* \in \arg \max_{m \in \{I, C, S\}} \text{LCB}[B_m(b)] \quad \text{subject to legality, stability, and cost.}$$

This makes Plan-style deployment a compiled output of the theory rather than a hand-written recipe. In a host that takes a hybrid approach similar to Qwen 3.6, GA ports often route to Stream-FGR because they are stateless gated-selection ports. GDN ports route to C-FGR or identity unless a finite-horizon certificate allows Stream-FGR.

5 Measurable Conditions for Frame Routing

5.1 Eligibility and Activation

Sections 3 and 4 define how a frame can be applied once a legal port has been chosen. The next question is more empirical: which ports should receive a frame at all? We separate two notions. A port is *eligible* if the host architecture exposes a legal signal coordinate. A port is *activated* only if its measured response is non-flat, its current Fisher is misaligned with that response, and its restricted gauge class has positive gain after confidence and cost checks. This distinction matters for hybrid hosts such as Qwen3.6-27B, where the architecture exposes repeated Gated DeltaNet and Gated Attention blocks [25]. Architecture gives candidate ports. Measurement decides whether Gyre acts.

The first diagnostic is response anisotropy. For a response Gramian $R \succeq 0$, define $\kappa_R = \frac{\lambda_{\max}(R)}{\text{tr}(R)/d}$. If κ_R is close to one, then the host treats all directions nearly equally in the measured response, and an orthogonal frame has little room to improve the Fisher-Gramian score. We therefore use non-flatness as an admission test, not as an assumption. A legal port with $\kappa_R \approx 1$ remains eligible, but the controller routes it to identity. This design prevents the paper from claiming that every exposed port contains useful routing structure.

Proposition 1 (Approximate flat-response abstention). *Let $F \succeq 0$ and let $R = \alpha I + E$ with $\|E\|_{\text{op}} \leq \epsilon_R$. Then for every $G \in O(d)$,*

$$|\text{tr}(G^\top FGR) - \text{tr}(FR)| \leq 2\epsilon_R \text{tr}(F).$$

This proposition gives a quantitative version of the flat-port rule. Exact flatness is rare in trained models, so the useful question is how far the measured response lies from a scalar matrix. When the deviation is small, any possible gain is bounded by a small multiple of the Fisher mass. The method can then spend its budget elsewhere.

5.2 Non-flatness of Gated Attention

Gated Attention is a plausible stateless host for frame routing because its output map contains learned selection structure. In the GA-Sel-FGR placement from Eq. (4.2), the response Gramian is

$$R_{\text{GA}}^{\text{sel}} = \mathbb{E}_t[D(g_t)^\top W_O^\top W_O D(g_t)].$$

This matrix can be non-flat for two independent reasons. The gate can assign different second moments to different heads or channels, and the output projection can have unequal block power. Gated-attention models are designed around non-uniform selection, for example through head-specific sigmoid gates [24]. Gyre does not rely on that design fact as a proof. It measures the induced $R_{\text{GA}}^{\text{sel}}$ or its block approximation and activates only if the measured response has anisotropy and positive restricted gain.

A useful approximation makes the condition transparent. Suppose heads form blocks, the output projection has approximately block-diagonal energy

$$W_O^\top W_O \approx \text{blockdiag}(a_1 I, \dots, a_H I),$$

and the gate is approximately headwise,

$$D(g_t) \approx \text{blockdiag}(g_{1,t} I, \dots, g_{H,t} I).$$

Then

$$R_{\text{GA}}^{\text{sel}} \approx \text{blockdiag} \left(a_h \mathbb{E}_t[g_{h,t}^2] I \right)_{h=1}^H.$$

The response is flat only if the products $a_h \mathbb{E}[g_{h,t}^2]$ are equal across heads. This gives a measured target rather than a rhetorical claim: GA non-flatness is the variance of gate-weighted output capacity across head blocks.

5.3 Non-flatness of Gated DeltaNet

For recurrent hosts, non-flatness means unequal finite-horizon retention or influence. Consider a legal GDN-style signal port

$$z_{t+1} = A_t z_t + B_t u_t.$$

A perturbation u at time t affects the state h steps later through $\Phi_{t+1 \rightarrow t+h} B_t u$, where Φ is the product of frozen recurrent transitions. The horizon- h response Gramian is

$$R_{\text{GDN}}(h) = \mathbb{E}_t \left[B_t^\top \Phi_{t+1 \rightarrow t+h}^\top \Phi_{t+1 \rightarrow t+h} B_t \right].$$

266 For any direction u ,

$$u^\top R_{\text{GDN}}(h)u = \mathbb{E}_t [\|\Phi_{t+1 \rightarrow t+h} B_t u\|_2^2].$$

267 Thus the eigenstructure of $R_{\text{GDN}}(h)$ directly measures which input directions persist or influence
 268 the future state at horizon h . Gated DeltaNet layers are designed with learned gating and delta-rule
 269 updates [38], so anisotropic retention is plausible. The theory still treats it as a measurement: Gyre
 270 estimates horizon or banded response profiles and abstains when they are flat.

271 A single horizon may be too narrow because different tasks require different memory timescales. We
 272 therefore also use weighted response mixtures

$$R_{\text{GDN}}^w = \sum_{h=1}^H w_h R_{\text{GDN}}(h), \quad w_h \geq 0, \quad \sum_h w_h = 1.$$

273 This gives a port-band view of retention. Short-horizon bands can support fast local adaptation, while
 274 long-horizon bands are candidates for C-FGR when their segment length justifies calibration or replay.
 275 Stream-FGR over recurrent bands remains gated by the stability certificate from Section 4.5.

276 5.4 Conditions of Non-flatness

277 A non-flat response says that the host has unequal slots. It does not say the current prompt needs a
 278 different slot assignment. The true activation condition is misalignment between F and R within the
 279 production gauge class:

$$\Delta_{\mathcal{G}} = \max_{G \in \mathcal{G}} \text{tr}(G^\top FGR) - \text{tr}(FR).$$

280 If F and R already align under identity, then even a strongly non-flat R gives no reason to rotate. This
 281 is why the controller uses both κ_R and $\Delta_{\mathcal{G}}$. Non-flatness finds unequal host resources. Restricted
 282 gain asks whether the current prompt places predictive mass in the wrong resources.

283 **Proposition 2** (Non-flat aligned ports abstain). *Let F and R be simultaneously diagonalizable with*
 284 *eigenvalues sorted in the same order. Then identity is a maximizer of $\text{tr}(G^\top FGR)$ over $O(d)$, even*
 285 *when R is non-flat.*

286 The proposition is useful because it prevents a common overclaim. The existence of gated heads or
 287 recurrent decay channels does not imply Gyre should act. The current prompt must also induce a
 288 Fisher geometry that is poorly assigned to those directions.

289 6 Empirical Validation

290 We evaluate Gyre by asking whether the Fisher-Gramian admission rule predicts model behavior.
 291 The primary metric is relative NLL reduction,

$$100 \cdot \frac{\text{NLL}_{\text{identity}} - \text{NLL}_{\text{method}}}{\text{NLL}_{\text{identity}}},$$

292 measured under teacher forcing on held-out continuation tokens. This metric is closest to the theory
 293 because $\Delta_{\mathcal{G}}$ is a local predictive surrogate, while task accuracy can mix routing quality with decoding
 294 and formatting effects.

295 Figure 2 shows the central validation. Restricted Fisher-Gramian gain predicts relative NLL reduction.
 296 The Gyre selector rises from near zero in low-gain bins to 1.9% in the top decile, while random gauges
 297 stay near zero and activation-only routing is much weaker. This pattern supports the mechanism:
 298 gains appear when the measured Fisher-response geometry predicts them, and they disappear under
 299 controls that remove Fisher alignment.

300 Table 1 summarizes quality and cost. The selector improves NLL by 0.86% on average and 1.90% on
 301 the top gain decile, with an 82.2% prompt win rate and a 1.6 point gain on selected long-context tasks.
 302 Its total wall-clock overhead is 7.0%, below the default 10% deployment budget. The diagnostic
 303 full-pass variant improves more, but costs 70.0% additional wall-clock time, so we treat it as an
 304 upper-bound diagnostic rather than a serving configuration.

305 Three additional checks support the same mechanism. An admission funnel from 4480 eligible
 306 port-prompt pairs to the 17.8% deployed under both geometric and cost gates, broken down by port
 307 family with matched-random-gauge contrasts, appears in Table 2 (Appendix F). The realized fraction
 308 $\eta_{\mathcal{G}}$ defined in Section 2 has port-family medians of 0.78, 0.71, 0.63, and 0.70 for GA-Sel, GA-Out,

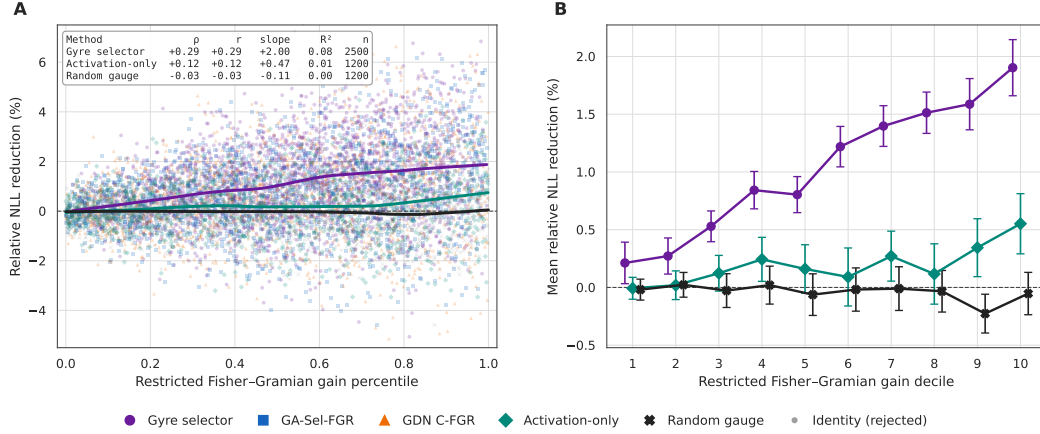


Figure 2: Restricted Fisher-Gramian gain predicts relative NLL reduction. Points show prompt-plan instances. Binned means show that Gyre improves as $\hat{\Delta}_G$ increases, while random gauges remain near identity and activation-only routing gives weaker gains.

Table 1: Frame routing improves selected prompts while preserving a small overhead budget.

Method	Deployed	Rel. NLL reduction $\uparrow$ (%)		Win rate	Task Δ	Wall-clk.	Decode tput.	Peak mem.
	(% elig.)	overall	top dec.	(%)	(pp)	$\Delta \downarrow$ (%)	Δ (%)	$\Delta \downarrow$ (%)
Identity	0.0	0.00	0.00	—	0.0	0.0	0.0	0.0
Random block gauge	17.8	-0.08 ± 0.06	-0.05 ± 0.18	49.0	-0.2	+2.6	-2.4	+0.6
Activation/PCA gauge	17.8	$+0.21 \pm 0.07$	$+0.55 \pm 0.26$	60.7	+0.3	+5.1	-4.7	+1.0
GA-only frame routing	11.2	$+0.55 \pm 0.07$	$+1.73 \pm 0.29$	78.2	+1.1	+3.3	-3.0	+0.7
GDN C-FGR (prefix)	9.6	$+0.45 \pm 0.08$	$+1.01 \pm 0.34$	68.8	+0.9	+10.6	-2.5	+1.5
Gyre selector	17.8	$+0.86 \pm 0.08$	$+1.90 \pm 0.24$	82.2	+1.6	+7.0	-4.2	+1.4
<i>Diagnostic upper bound, not a deployment mode (gray reference in Figure 3):</i>								
Diagnostic full-pass [†]	22.0	$+1.30 \pm 0.10$	$+3.60 \pm 0.32$	86.5	+2.4	+70.0	-32.6	+4.1

See Appendix F.6 for metric definitions, confidence intervals, controls, and diagnostic-row details.

GDN-short, and GDN-long respectively (Figure 5), showing that production gauges capture most of the full-frame oracle gain. Length-stratified results (Figure 6, Table 3) show an inverted-U peak around 32K on both NLL and state-tracking tasks, consistent with overhead dominance at short lengths and stale-Fisher effects at extreme lengths. Appendix F reports the quality-efficiency frontier, non-flatness funnel, mechanism audit, length scaling, dataset details, and training protocol.

Limitations Recurrent Stream-FGR is admitted only with a finite-horizon stability certificate, which our current measurements provide only for short bands; longer-horizon recurrent streaming therefore remains out of scope. At extreme context lengths ($\geq 128K$) prompt-local Fisher becomes stale relative to held-out suffixes and gains shrink, indicating that the calibration window itself is a design variable we have not yet optimized. Finally, all reported overheads are measured against a single serving stack; absolute wall-clock and memory deltas will shift with kernel fusion, paging, and quantization choices that lie outside the frame-routing layer.

7 Conclusion

Gyre frames test-time adaptation as coordinate allocation inside a frozen host. Our main claim is that non-flat retention and selection directions create a measurable routing opportunity when prompt-local Fisher is misaligned with host response. Fisher-Gramian routing gives the geometry, legal-port constraints prevent circular interventions, and the controller selects identity, C-FGR, or Stream-FGR under gain and cost constraints. Experiments on Qwen3.6-27B support the mechanism as restricted gain predicts NLL reduction, random gauges fail, and the selector improves quality under a small overhead budget. The broader lesson is simple, frozen models can adapt efficiently by changing frames without changing weights.

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A World-feedback Perspective

Gyre can also be viewed as feedback-conditioned control for frozen foundation-model inference. The relevant feedback is produced by the model and its digital execution environment, rather than by human preference labels. At calibration time, Gyre observes prompt-local loss sensitivity, host-response anisotropy, and restricted Fisher-Gramian gain. At deployment time, the controller also accounts for measurable system outcomes, including wall-clock overhead, decode throughput, peak memory, and a fixed serving budget. These signals determine whether an internal routing action is admitted.

For each legal port-band, the controller selects identity, C-FGR, or Stream-FGR subject to legality, state coherence, stability, and cost checks. The selected action changes the frame through which signal directions enter a frozen host resource. The base model weights, recurrence rules, masks, gates, and cache coordinates remain fixed under the legal-port contract. Thus, the feedback loop acts on inference-time routing decisions, while the underlying sequence model remains unchanged.

Gyre does not optimize long-horizon social utility or train a new reward model but instead asks when a prompt-local predictive direction is poorly assigned to a non-uniform internal resource, then accepts a frame only when the measured quality-cost evidence justifies the intervention. This gives a controlled setting for studying how foundation-model inference can use outcome-facing feedback from a digital environment.

The empirical protocol reflects this scope. Relative NLL reduction measures whether the Fisher-Gramian signal has value for predictive quality. Task deltas test whether the selected routing actions affect long-context behavior beyond the surrogate. Wall-clock, throughput, and memory measurements test whether the same actions remain usable under a serving budget. Random-gauge and activation-only controls separate arbitrary perturbation and unsupervised activation geometry from Fisher-response alignment.

Under this view, Gyre contributes a feedback interface for foundation-model inference. Internal predictive feedback proposes a routing action, host-response feedback limits where the action is meaningful, and deployment feedback decides whether the action is worth taking.

B Appendix for Section 2: Mathematical Foundations

B.1 Frozen-host Setup

We formalize the objects used in Section 2. The appendix uses a slightly more explicit notation than the main text because several apparent variants of frame routing become invalid once control variables, cache coordinates, or recurrent state coherence are included.

Definition 1 (Frozen sequence host). *A frozen sequence host is an autoregressive model with parameters θ_0 fixed during inference. For a prefix $x_{\leq t}$, the host produces a distribution $p_{\theta_0}(\cdot \mid x_{\leq t})$ and next-token loss*

$$\ell_t(\theta_0) = -\log p_{\theta_0}(x_{t+1} \mid x_{\leq t}).$$

Any test-time operation considered in this paper may alter intermediate coordinates but may not update θ_0 .

Definition 2 (Port and control variables). *A port p exposes an internal signal coordinate*

$$u_{t,p} \in \mathbb{R}^{d_p}$$

inside the frozen host computation. Locally, the host computation around the port is written

$$H_{t,p}(u_{t,p}; c_{t,p}),$$

where $c_{t,p}$ denotes control variables such as gates, masks, attention logits, recurrence coefficients, routers, cache addresses, or decay parameters.

Assumption 1 (Signal-control separation). *A Gyre frame at port p acts only on the signal coordinate:*

$$u_{t,p} \mapsto G_p^\top u_{t,p}, \quad G_p \in O(d_p).$$

The control variables $c_{t,p}$ are computed by the frozen host and are not changed by the candidate frame.

573 **Remark 1** (Why signal-control separation is required). *The Fisher-Gramian objective treats the host*
 574 *response Gramian as a fixed matrix. If the candidate frame also changes gates, decay coefficients,*
 575 *routing logits, attention masks, or cache coordinates, then the response Gramian becomes a function*
 576 *$R(G)$. The objective would then be*

$$G \mapsto \text{tr}(G^\top FGR(G)),$$

577 *which no longer admits the alignment result in Theorem 1. The legal-port contract is therefore part*
 578 *of the mathematics, not an implementation preference.*

579 **B.2 Predictive Fisher and Response Gramian**

580 **Definition 3** (Prompt-local predictive Fisher). *For a port p , let*

$$g_{t,p} = \nabla_{u_{t,p}} \ell_t(\theta_0).$$

581 *Given a calibration set $\mathcal{T}$ of positions, the empirical prompt-local Fisher is*

$$\hat{F}_p = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} g_{t,p} g_{t,p}^\top.$$

582 *The population version is denoted*

$$F_p = \mathbb{E}_t[g_{t,p} g_{t,p}^\top].$$

583 **Definition 4** (Host response map and Gramian). *Let $r_{t,p}$ denote the downstream host quantity whose*
 584 *selected or retained energy we care about. For a stateless port, $r_{t,p}$ can be the local block output.*
 585 *For a recurrent port, $r_{t,p}$ can be a finite-horizon state or output response. Let*

$$J_{t,p} = \frac{\partial r_{t,p}}{\partial u_{t,p}}$$

586 *be the local response Jacobian under the frozen host policy. The response Gramian is*

$$R_p = \mathbb{E}_t[J_{t,p}^\top J_{t,p}],$$

587 *or empirically*

$$\hat{R}_p = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} J_{t,p}^\top J_{t,p}.$$

588 **Lemma 1** (Positive semidefiniteness). *For every port p , both F_p and R_p are positive semidefinite.*

589 *Proof.* For any $a \in \mathbb{R}^{d_p}$,

$$a^\top F_p a = \mathbb{E}_t[a^\top g_{t,p} g_{t,p}^\top a] = \mathbb{E}_t[(g_{t,p}^\top a)^2] \geq 0.$$

590 Similarly,

$$a^\top R_p a = \mathbb{E}_t[a^\top J_{t,p}^\top J_{t,p} a] = \mathbb{E}_t[\|J_{t,p} a\|_2^2] \geq 0.$$

591 □

592 **Definition 5** (Fisher-Gramian score). *For $F, R \succeq 0$ and $G \in O(d)$, define*

$$\mathcal{J}(G; F, R) = \text{tr}(G^\top FGR).$$

593 *When F, R are clear from context, we write $\mathcal{J}(G)$.*

594 **Interpretation 1** (Direction-to-slot assignment). *The matrix F measures which directions are pre-*
 595 *dictive for the current calibration set. The matrix R measures which host directions are selected or*
 596 *retained by the frozen computation. The score $\mathcal{J}(G)$ measures how much predictive mass lands in*
 597 *high-response directions after the frame G is applied.*

598 B.3 Full-frame Alignment

599 **Theorem 3** (Full-frame Fisher-Gramian alignment). *Let $F, R \succeq 0$ have eigendecompositions*

$$F = U_F \text{diag}(\sigma_1, \dots, \sigma_d) U_F^\top, \quad R = U_R \text{diag}(r_1, \dots, r_d) U_R^\top,$$

600 *where*

$$\sigma_1 \geq \dots \geq \sigma_d \geq 0, \quad r_1 \geq \dots \geq r_d \geq 0.$$

601 *Then*

$$\max_{G \in O(d)} \text{tr}(G^\top F G R) = \sum_{i=1}^d \sigma_i r_i.$$

602 *One maximizer is*

$$G^\star = U_F U_R^\top.$$

603 *More generally, any maximizer aligns eigenspaces of R with eigenspaces of F in sorted order, with*
 604 *arbitrary orthogonal rotations inside equal-eigenvalue subspaces.*

605 *Proof.* Let

$$M = U_F^\top G U_R.$$

606 Since G, U_F, U_R are orthogonal, $M \in O(d)$. By cyclicity of trace,

$$\text{tr}(G^\top F G R) = \text{tr}(M^\top \text{diag}(\sigma_i) M \text{diag}(r_j)).$$

607 Expanding gives

$$\text{tr}(G^\top F G R) = \sum_{i,j} \sigma_i r_j M_{ij}^2.$$

608 Define $P_{ij} = M_{ij}^2$. Because M is orthogonal,

$$\sum_j P_{ij} = 1, \quad \sum_i P_{ij} = 1,$$

609 so P is doubly stochastic. The Birkhoff-von Neumann theorem states that every doubly stochastic
 610 matrix is a convex combination of permutation matrices. Hence the maximum of the linear function

$$P \mapsto \sum_{i,j} \sigma_i r_j P_{ij}$$

611 over the doubly stochastic polytope is attained at a permutation matrix. Since both sequences (σ_i)
 612 and (r_i) are sorted in descending order, the rearrangement inequality gives

$$\sum_{i,j} \sigma_i r_j P_{ij} \leq \sum_i \sigma_i r_i.$$

613 This upper bound is achieved by $M = I$, equivalently $G = U_F U_R^\top$. □

614 **Corollary 1** (Flat-response abstention). *If $R = \alpha I$ for some $\alpha \geq 0$, then*

$$\mathcal{J}(G) = \alpha \text{tr}(F)$$

615 *for every $G \in O(d)$.*

Proof.

$$\mathcal{J}(G) = \text{tr}(G^\top F G \alpha I) = \alpha \text{tr}(G^\top F G) = \alpha \text{tr}(F).$$

616 □

617 **Corollary 2** (Aligned identity). *If F and R share an eigenbasis with eigenvalues sorted in the same*
 618 *order, then $G = I$ is a full-frame maximizer.*

619 *Proof.* If $F = U \text{diag}(\sigma_i) U^\top$ and $R = U \text{diag}(r_i) U^\top$ with both spectra sorted in the same order,
 620 the identity frame already aligns the eigenspaces required by Theorem 3. □

621 B.4 Restricted Production Gauges

622 **Definition 6** (Restricted gauge class). *A production gauge class is a subset*

$$\mathcal{G} \subseteq O(d)$$

623 *chosen for computational, hardware, stability, or cache reasons. Examples include block permutations,*
 624 *block Givens rotations, block-diagonal orthogonal matrices, head-wise rotations, Householder*
 625 *products, and quantization-group-aligned rotations.*

626 **Definition 7** (Oracle and restricted gains). *The full-frame oracle gain is*

$$\Delta_{O(d)} = \max_{G \in O(d)} \mathcal{J}(G) - \mathcal{J}(I).$$

627 *The restricted gain is*

$$\Delta_{\mathcal{G}} = \max_{G \in \mathcal{G}} \mathcal{J}(G) - \mathcal{J}(I).$$

628 *If $\Delta_{O(d)} > 0$, the realized fraction is*

$$\eta_{\mathcal{G}} = \frac{\Delta_{\mathcal{G}}}{\Delta_{O(d)} + \varepsilon}$$

629 *for a small numerical $\varepsilon > 0$.*

630 **Proposition 3** (Restricted gain is bounded by oracle gain). *If $\mathcal{G} \subseteq O(d)$ and $I \in \mathcal{G}$, then*

$$0 \leq \Delta_{\mathcal{G}} \leq \Delta_{O(d)}.$$

631 *Proof.* Because $I \in \mathcal{G}$,

$$\max_{G \in \mathcal{G}} \mathcal{J}(G) \geq \mathcal{J}(I),$$

632 so $\Delta_{\mathcal{G}} \geq 0$. Since $\mathcal{G} \subseteq O(d)$,

$$\max_{G \in \mathcal{G}} \mathcal{J}(G) \leq \max_{G \in O(d)} \mathcal{J}(G),$$

633 which gives the upper bound. □

634 **Definition 8** (Candidate gain). *In practice we may use a candidate $\hat{G} \in \mathcal{G}$ produced by an optimizer*
 635 *or heuristic. Its candidate gain is*

$$\hat{\Delta}(\hat{G}) = \mathcal{J}(\hat{G}) - \mathcal{J}(I).$$

636 *The runtime admission rule may reject a candidate if $\hat{\Delta}(\hat{G}) \leq \tau_{\Delta}$.*

637 **Remark 2** (Exact gain versus estimated gain). *The exact restricted gain $\Delta_{\mathcal{G}}$ is nonnegative when*
 638 *$I \in \mathcal{G}$. A computed candidate can have negative or statistically uncertain gain. The controller*
 639 *therefore uses candidate gain or a confidence lower bound, while the theory uses $\Delta_{\mathcal{G}}$ for the ideal*
 640 *restricted class.*

641 B.5 Finite-sample Estimates and Admission Confidence

642 **Definition 9** (Estimated Fisher-Gramian score). *Given empirical estimates $\hat{F}, \hat{R}$, define*

$$\hat{\mathcal{J}}(G) = \text{tr}(G^{\top} \hat{F} G \hat{R}).$$

643 **Lemma 2** (Score perturbation bound). *Let $F, R, \hat{F}, \hat{R} \succeq 0$. For any $G \in O(d)$,*

$$|\hat{\mathcal{J}}(G) - \mathcal{J}(G)| \leq \|F - \hat{F}\|_{\text{F}} \|R\|_{\text{F}} + \|\hat{F}\|_{\text{F}} \|R - \hat{R}\|_{\text{F}}.$$

644 *Proof.* Write

$$\hat{\mathcal{J}}(G) - \mathcal{J}(G) = \text{tr}(G^{\top} (\hat{F} - F) G R) + \text{tr}(G^{\top} \hat{F} G (\hat{R} - R)).$$

645 Using Cauchy-Schwarz for the Frobenius inner product,

$$|\text{tr}(A^{\top} B)| \leq \|A\|_{\text{F}} \|B\|_{\text{F}}.$$

646 Orthogonal conjugation preserves Frobenius norm, so

$$\|G^{\top} (\hat{F} - F) G\|_{\text{F}} = \|\hat{F} - F\|_{\text{F}},$$

647 and

$$\|G^{\top} \hat{F} G\|_{\text{F}} = \|\hat{F}\|_{\text{F}}.$$

648 Combining the two terms proves the claim. □

649 **Corollary 3** (Conservative admission with estimation error). *If a candidate $\hat{G}$ satisfies*

$$\hat{\mathcal{J}}(\hat{G}) - \hat{\mathcal{J}}(I) > 2\epsilon_{\text{est}},$$

650 *and the uniform estimation error obeys*

$$\sup_{G \in \{I, \hat{G}\}} |\hat{\mathcal{J}}(G) - \mathcal{J}(G)| \leq \epsilon_{\text{est}},$$

651 *then*

$$\mathcal{J}(\hat{G}) > \mathcal{J}(I).$$

Proof.

$$\mathcal{J}(\hat{G}) - \mathcal{J}(I) \geq \hat{\mathcal{J}}(\hat{G}) - \hat{\mathcal{J}}(I) - 2\epsilon_{\text{est}} > 0.$$

652 □

653 B.6 Relation Between Surrogate Gain and Loss

654 **Remark 3** (What the Fisher-Gramian score proves). *The Fisher-Gramian score is a local routing*
 655 *surrogate. The alignment theorem proves optimality for this surrogate under fixed F, R . It does not*
 656 *alone prove that next-token loss decreases. The paper’s empirical burden is therefore to test whether*
 657 *$\Delta_{\mathcal{G}}$ predicts changes in next-token loss or task metrics across prompts and legal ports.*

658 **Definition 10** (Empirical validation target). *For a calibration or evaluation set, the key validation*
 659 *relation is*

$$\Delta_{\mathcal{G}}(p) \quad \text{versus} \quad -\Delta_{\text{NLL}}(p)$$

660 *across prompts or prompt-port pairs p . A strong theory-driven experiment checks whether positive*
 661 *restricted gain predicts lower next-token loss, while flat ports, invalid ports, and random gauges*
 662 *behave like controls.*

663 C Appendix for Sections 3 and 4

664 C.1 Objects and Assumptions

665 **Definition 11** (Frozen host policy). *A frozen host policy is the computation graph induced by the*
 666 *fixed parameters θ_0 together with all control variables computed from the input prefix, including*
 667 *attention masks, gates, decay parameters, recurrence coefficients, routers, and cache addresses. A*
 668 *Gyre intervention may alter signal coordinates but may not alter the rule that computes these control*
 669 *variables.*

670 **Definition 12** (Legal signal port). *A port p is legal if the local host computation can be written as*

$$H_{t,p}(u_{t,p}; c_{t,p}),$$

671 *where $u_{t,p} \in \mathbb{R}^{d_p}$ is a signal coordinate and $c_{t,p}$ is fixed under candidate gauges. A legal Gyre frame*
 672 *has the form*

$$u_{t,p} \mapsto G^\top u_{t,p}, \quad G \in O(d_p).$$

673 **Assumption 2** (Fixed-response assumption). *For a legal port, the response Gramian*

$$R = \mathbb{E}_t[J_t^\top J_t]$$

674 *is measured under the frozen host policy and does not depend on the candidate gauge G .*

675 **Assumption 3** (Restricted gauge class). *Each deployed port uses a stated production gauge class*

$$\mathcal{G} \subseteq O(d)$$

676 *that contains the identity. The class may impose block, head, quantization-group, or low-rank*
 677 *structure.*

678 **Definition 13** (Candidate and restricted gain). *For fixed $F, R \succeq 0$, define*

$$\mathcal{J}(G) = \text{tr}(G^\top F G R).$$

679 *For a candidate $\hat{G} \in \mathcal{G}$, define*

$$\hat{\Delta}(\hat{G}) = \mathcal{J}(\hat{G}) - \mathcal{J}(I).$$

680 *The ideal restricted gain is*

$$\Delta_{\mathcal{G}} = \max_{G \in \mathcal{G}} \mathcal{J}(G) - \mathcal{J}(I).$$

681 C.2 Admission Theorem

682 **Theorem 4** (Exact restricted admission). *Under Assumption 3, let*

$$G_G^* \in \arg \max_{G \in \mathcal{G}} \mathcal{J}(G).$$

683 *Then*

$$\mathcal{J}(G_G^*) - \mathcal{J}(I) = \Delta_G.$$

684 *Identity is a restricted maximizer if and only if $\Delta_G = 0$.*

685 *Proof.* The first statement follows by substituting the maximizer into the definition of Δ_G . If $\Delta_G = 0$,
686 then the maximum restricted score equals $\mathcal{J}(I)$, so identity is a maximizer. If identity is a maximizer,
687 then the maximum restricted score is $\mathcal{J}(I)$, so $\Delta_G = 0$. $\square$

688 **Corollary 4** (Flat-response rejection). *If $R = \alpha I$ for $\alpha \geq 0$, then $\Delta_G = 0$ for every $\mathcal{G} \subseteq O(d)$*
689 *containing identity.*

690 *Proof.* For any $G \in O(d)$,

$$\mathcal{J}(G) = \text{tr}(G^\top F G \alpha I) = \alpha \text{tr}(G^\top F G) = \alpha \text{tr}(F) = \mathcal{J}(I).$$

691 Thus every restricted gauge has the same score. $\square$

692 **Lemma 3** (Estimated-score perturbation). *Let $\hat{F}, \hat{R}$ be estimates of F, R . For any $G \in O(d)$,*

$$\left| \text{tr}(G^\top \hat{F} G \hat{R}) - \text{tr}(G^\top F G R) \right| \leq \|\hat{F} - F\|_F \|R\|_F + \|\hat{F}\|_F \|\hat{R} - R\|_F.$$

693 *Proof.* Write the difference as

$$\text{tr}(G^\top (\hat{F} - F) G R) + \text{tr}(G^\top \hat{F} G (\hat{R} - R)).$$

694 Cauchy-Schwarz for the Frobenius inner product gives

$$|\text{tr}(A^\top B)| \leq \|A\|_F \|B\|_F.$$

695 Orthogonal conjugation preserves Frobenius norm, which yields the result. $\square$

696 **Corollary 5** (Conservative lower-confidence admission). *Suppose a candidate $\hat{G}$ satisfies*

$$\hat{\mathcal{J}}(\hat{G}) - \hat{\mathcal{J}}(I) > 2\epsilon$$

697 *and both candidate scores have estimation error at most ϵ . Then*

$$\mathcal{J}(\hat{G}) > \mathcal{J}(I).$$

Proof.

$$\mathcal{J}(\hat{G}) - \mathcal{J}(I) \geq \hat{\mathcal{J}}(\hat{G}) - \hat{\mathcal{J}}(I) - 2\epsilon > 0.$$

698 $\square$

699 C.3 Attention-side Ports

700 **Definition 14** (Gated-attention host). *A gated-attention output has the form*

$$o_t = \text{SDPA}(Q, K, V)_t, \quad y_t = W_O D(g_t) o_t,$$

701 *where $D(g_t)$ is diagonal.*

702 **Proposition 4** (GA-Sel-FGR Gramian). *For*

$$\tilde{y}_t = W_O D(g_t) G^\top o_t,$$

703 *the response Gramian with respect to $G^\top o_t$ is*

$$R_{\text{GA}}^{\text{sel}} = \mathbb{E}_t[D(g_t)^\top W_O^\top W_O D(g_t)].$$

704 *Proof.* Let $v_t = G^\top o_t$. The local map from v_t to output is

$$v_t \mapsto W_O D(g_t) v_t.$$

705 The Jacobian is $J_t = W_O D(g_t)$. Thus

$$J_t^\top J_t = D(g_t)^\top W_O^\top W_O D(g_t).$$

706 Taking expectation proves the claim. $\square$

707 **Proposition 5** (GA-Out-FGR Gramian). *For*

$$h_t = D(g_t) o_t, \quad \tilde{y}_t = W_O G^\top h_t,$$

708 *the response Gramian with respect to $G^\top h_t$ is*

$$R_{\text{GA}}^{\text{out}} = W_O^\top W_O.$$

709 *Proof.* Let $v_t = G^\top h_t$. The local map is $v_t \mapsto W_O v_t$, whose Jacobian is W_O . Hence $J_t^\top J_t =$
710 $W_O^\top W_O$. $\square$

711 **Corollary 6** (Placement-dependent Gramians). *GA-Sel-FGR and GA-Out-FGR generally induce*
712 *different response Gramians. They coincide only under special gate structures, for example constant*
713 *scalar gates up to scale.*

714 *Proof.* The result follows from Propositions 4 and 5. For channel-specific or head-specific gates,
715 $D(g_t)$ does not commute away as a scalar, so the two matrices differ in general. $\square$

716 **Lemma 4** (Symmetric QK rotation is logit-invariant). *If $G \in O(d)$, $Q' = QG$, and $K' = KG$, then*

$$Q' K'^\top = Q K^\top.$$

Proof.

$$Q' K'^\top = QG(KG)^\top = QGG^\top K^\top = QK^\top.$$

717 $\square$

718 **Proposition 6** (Cache coherence for value-side streaming). *Suppose a value cache stores time-*
719 *varying framed entries $G_i^\top v_i$. A later read that combines them as if they share one coordinate system*
720 *is equivalent to a valid common-frame read for all possible values only if all active frames agree on*
721 *the read subspace, or if the cache carries frame metadata and converts entries before combination.*

722 *Proof.* A common-frame read has the form

$$G_\star^\top \sum_i a_i v_i.$$

723 A time-varying framed read gives

$$\sum_i a_i G_i^\top v_i.$$

724 For these to agree for arbitrary a_i and v_i , each $G_i^\top$ must equal $G_\star^\top$ on the active value subspace.
725 Otherwise two choices of v_i supported on a direction where the frames differ produce different
726 reads. Metadata or replay restores coherence by converting entries into a common frame before
727 summation. $\square$

728 **C.4 Recurrent C-FGR**

729 **Definition 15** (Legal recurrent signal port). *A recurrent port is legal when the frozen recurrence can*
 730 *be written locally as*

$$z_{t+1} = A_t z_t + B_t u_t$$

731 *and a Gyre frame changes only the signal write:*

$$z_{t+1} = A_t z_t + B_t G^\top u_t.$$

732 *The computation of A_t and B_t may be input-dependent, but it is fixed with respect to the candidate*
 733 *gauge.*

734 **Theorem 5** (C-FGR segment coherence). *For a fixed segment gauge G_s and recurrence*

$$z_{t+1} = A_t z_t + B_t G_s^\top u_t,$$

735 *define*

$$\Phi_{a \rightarrow b} = A_{b-1} A_{b-2} \cdots A_a$$

736 *for $a < b$, and $\Phi_{b \rightarrow b} = I$. Then*

$$z_T = \Phi_{0 \rightarrow T} z_0 + \sum_{t=0}^{T-1} \Phi_{t+1 \rightarrow T} B_t G_s^\top u_t.$$

737 *All signal writes in the segment use the same frame.*

738 *Proof.* The proof is by induction. The claim is immediate for $T = 1$. If it holds at T , then

$$z_{T+1} = A_T z_T + B_T G_s^\top u_T$$

739

$$= A_T \Phi_{0 \rightarrow T} z_0 + \sum_{t=0}^{T-1} A_T \Phi_{t+1 \rightarrow T} B_t G_s^\top u_t + B_T G_s^\top u_T.$$

740 Using $A_T \Phi_{a \rightarrow T} = \Phi_{a \rightarrow T+1}$ gives the desired form at $T + 1$. □

741 **Corollary 7** (Valid C-FGR segment contracts). *C-FGR is coherent when one of the following holds:*
 742 *the gauge is static before the segment begins, the segment is replayed after fitting the gauge, the claim*
 743 *is restricted to future writes after a prefix boundary, or the recurrent state is reset before applying the*
 744 *gauge.*

745 *Proof.* Each listed contract makes the recurrence inside the claimed segment match the fixed-gauge
 746 recurrence in Theorem 5. □

747 **Proposition 7** (Mid-state switching mixes frames). *If positions $0, \dots, S - 1$ are written under*
 748 *identity and positions $S, \dots, T - 1$ are written under G_s without replay, then*

$$z_T = \Phi_{0 \rightarrow T} z_0 + \sum_{t=0}^{S-1} \Phi_{t+1 \rightarrow T} B_t u_t + \sum_{t=S}^{T-1} \Phi_{t+1 \rightarrow T} B_t G_s^\top u_t.$$

749 *Unless G_s equals identity on the active signal subspace, the state contains mixed frames.*

750 *Proof.* The expression follows by unrolling the recurrence before and after the switch. The first sum
 751 contains identity-frame writes, while the second contains G_s -frame writes. □

752 **C.5 Stream-FGR and Recurrent Drift**

753 **Definition 16** (Stream-FGR curve). *A Stream-FGR curve is a sequence*

$$G_{t+1} = G_t \exp(\Omega_t), \quad \Omega_t^\top = -\Omega_t.$$

754 **Lemma 5** (Orthogonality preservation). *If $G_t \in O(d)$ and $\Omega_t^\top = -\Omega_t$, then $G_{t+1} \in O(d)$.*

755 *Proof.* For skew-symmetric Ω_t ,

$$\exp(\Omega_t)^\top = \exp(-\Omega_t) = \exp(\Omega_t)^{-1}.$$

756 Thus $\exp(\Omega_t)$ is orthogonal, and the product of orthogonal matrices is orthogonal. $\square$

757 **Assumption 4** (Finite-horizon response bound). *For a recurrent host over horizon H ,*

$$\|\Phi_{t+1 \rightarrow T}\|_2 \leq K\rho^{T-t-1}$$

758 *for all $T - t \leq H$, with $K \geq 1$ and $\rho \geq 0$.*

759 **Assumption 5** (Bounded write and signal). *For the same horizon,*

$$\|B_t\|_2 \leq B, \quad \|u_t\|_2 \leq U.$$

760 **Assumption 6** (Bounded frame speed). *The stream satisfies*

$$\|G_{t+1} - G_t\|_2 \leq \eta$$

761 *at every step.*

762 **Lemma 6** (Frame-distance accumulation). *Under Assumption 6,*

$$\|G_t - G_T\|_2 \leq (T - t)\eta.$$

763 *Proof.* Apply the triangle inequality:

$$\|G_t - G_T\|_2 \leq \sum_{s=t}^{T-1} \|G_s - G_{s+1}\|_2 \leq (T - t)\eta.$$

764 $\square$

765 **Theorem 6** (Finite-horizon recurrent Stream-FGR drift). *Let*

$$z_T = \sum_{t=T-H}^{T-1} \Phi_{t+1 \rightarrow T} B_t G_t^\top u_t$$

766 *and*

$$z_T^* = \sum_{t=T-H}^{T-1} \Phi_{t+1 \rightarrow T} B_t G_T^\top u_t.$$

767 *Under Assumptions 4, 5, and 6,*

$$\|z_T - z_T^*\|_2 \leq KBU\eta \sum_{h=0}^{H-1} (h+1)\rho^h.$$

768 *For $\rho \neq 1$,*

$$\sum_{h=0}^{H-1} (h+1)\rho^h = \frac{1 - (H+1)\rho^H + H\rho^{H+1}}{(1-\rho)^2}.$$

769 *For $\rho = 1$, the sum equals $H(H+1)/2$.*

770 *Proof.* Subtract:

$$z_T - z_T^* = \sum_{t=T-H}^{T-1} \Phi_{t+1 \rightarrow T} B_t (G_t^\top - G_T^\top) u_t.$$

771 Taking norms and applying submultiplicativity gives

$$\|z_T - z_T^*\|_2 \leq \sum_{t=T-H}^{T-1} \|\Phi_{t+1 \rightarrow T}\|_2 \|B_t\|_2 \|G_t - G_T\|_2 \|u_t\|_2.$$

772 Using the three assumptions and Lemma 6,

$$\|z_T - z_T^*\|_2 \leq \sum_{t=T-H}^{T-1} K \rho^{T-t-1} B U (T-t) \eta.$$

773 Set $h = T - t - 1$. Then $T - t = h + 1$, so the sum becomes

$$K B U \eta \sum_{h=0}^{H-1} (h+1) \rho^h.$$

774 The closed forms follow from the finite derivative of the geometric series. $\square$

775 **Corollary 8** (Stable tail and velocity budget). *If $\rho < 1$ and an infinite-tail approximation is used,*
776 *then*

$$\|z_T - z_T^*\|_2 \leq \frac{K B U \eta}{(1 - \rho)^2}.$$

777 To keep this bound below ϵ , it suffices to use

$$\eta \leq \frac{\epsilon(1 - \rho)^2}{K B U}.$$

778 For a finite horizon H , it suffices to use

$$\eta \leq \frac{\epsilon}{K B U \sum_{h=0}^{H-1} (h+1) \rho^h}.$$

779 *Proof.* For $\rho < 1$,

$$\sum_{h=0}^{\infty} (h+1) \rho^h = \frac{1}{(1 - \rho)^2}.$$

780 Rearranging the finite and infinite bounds gives the velocity budgets. $\square$

781 **Remark 4** (No default $K = 1$). *The factor K captures finite-horizon transient amplification. It*
782 *equals one only under a certified response bound, such as a normal contraction in the measured norm.*
783 *Recurrent hosts with non-normal dynamics can have large transient growth, so a small spectral*
784 *radius alone does not justify $K = 1$.*

785 **Corollary 9** (Stateless ports lack recurrent frame mixing). *For a stateless output port whose output*
786 *at time t depends only on $G_t^\top u_t$, the recurrent drift term in Theorem 6 does not arise.*

787 *Proof.* The theorem bounds error from accumulating writes made under different frames inside a
788 state. A stateless port has no such accumulation. Each position is evaluated in its current frame. $\square$

789 C.6 Mode Utility and Regime Behavior

790 **Definition 17** (Mode utilities). *For a port-band over a segment of expected length S , let identity*
791 *have utility*

$$B_I(S) = 0.$$

792 Let C-FGR have gain Δ_C , fixed cost C_A , and risk penalty R_C :

$$B_C(S) = \Delta_C S - C_A - R_C.$$

793 Let Stream-FGR have gain Δ_S , time-to-target τ , per-token cost c_S , and risk penalty R_S :

$$B_S(S) = \begin{cases} \frac{\Delta_S S^2}{2\tau} - c_S S - R_S, & S \leq \tau, \\ \Delta_S(S - \tau/2) - c_S S - R_S, & S > \tau. \end{cases}$$

794 **Proposition 8** (Short-context identity regime). *If $C_A + R_C > 0$ and either $R_S > 0$ or $c_S > 0$, then*
 795 *there exists $S_0 > 0$ such that for all $0 < S < S_0$,*

$$B_C(S) < 0, \quad B_S(S) < 0.$$

796 Thus identity has maximal utility among these three modes for sufficiently short segments.

797 *Proof.* As $S \rightarrow 0$,

$$B_C(S) \rightarrow -C_A - R_C < 0.$$

798 For Stream-FGR with $S \leq \tau$,

$$B_S(S) = \frac{\Delta_S S^2}{2\tau} - c_S S - R_S.$$

799 If $R_S > 0$, this tends to $-R_S < 0$. If $R_S = 0$ and $c_S > 0$, the linear negative term dominates the
 800 quadratic positive term for sufficiently small S . Hence both non-identity modes are negative on a
 801 small interval. $\square$

802 **Proposition 9** (C-FGR long-segment amortization). *If $\Delta_C > 0$, then*

$$B_C(S) > 0$$

803 whenever

$$S > \frac{C_A + R_C}{\Delta_C}.$$

804 *Proof.* Substitute into

$$B_C(S) = \Delta_C S - C_A - R_C.$$

805 $\square$

806 **Proposition 10** (Stream-FGR after target is reached). *For $S > \tau$, Stream-FGR beats identity if*

$$(\Delta_S - c_S)S > \frac{\Delta_S \tau}{2} + R_S.$$

807 *Proof.* For $S > \tau$,

$$B_S(S) = \Delta_S(S - \tau/2) - c_S S - R_S.$$

808 The condition $B_S(S) > 0$ rearranges to the stated inequality. $\square$

809 **Proposition 11** (When C-FGR beats Stream-FGR for long recurrent segments). *For $S > \tau$, C-FGR*
 810 *beats Stream-FGR when*

$$(\Delta_C - \Delta_S + c_S)S > C_A + R_C - R_S - \frac{\Delta_S \tau}{2}.$$

811 *Proof.* Use $B_C(S) > B_S(S)$:

$$\Delta_C S - C_A - R_C > \Delta_S(S - \tau/2) - c_S S - R_S.$$

812 Collect terms in S . $\square$

813 **Remark 5** (Regime interpretation). *The utility model explains why identity often wins in short*
 814 *contexts, Stream-FGR can win in medium contexts when τ and risk are small, and C-FGR can win*
 815 *in long recurrent contexts when fixed costs amortize and streaming carries a state-mixing penalty.*
 816 *These are regime claims. The controller uses measured gain, cost, and stability quantities rather than*
 817 *hard-coding the regimes.*

818 C.7 Runtime Selection

819 **Definition 18** (Feasible mode set). *For each port-band b , define*

$$\mathcal{M}_b \subseteq \{I, C, S\}$$

820 *as the modes satisfying the legal-port contract, C-FGR segment contract, Stream-FGR stability*
821 *certificate, cache contract, and latency/memory constraints.*

822 **Definition 19** (Lower-confidence utility). *Let $\hat{B}_m(b)$ be the estimated utility of mode m on band b ,*
823 *and let $\text{rad}_m(b)$ be a confidence radius. Define*

$$\text{LCB}_m(b) = \hat{B}_m(b) - \text{rad}_m(b).$$

824 **Proposition 12** (Feasible lower-confidence routing). *The controller*

$$m_b^* \in \arg \max_{m \in \mathcal{M}_b} \text{LCB}_m(b)$$

825 *selects a feasible mode. If identity has the largest lower-confidence utility, the controller abstains.*

826 *Proof.* The maximization domain is exactly the feasible set $\mathcal{M}_b$, so the selected mode is feasible.
827 Since identity belongs to $\mathcal{M}_b$, if its lower-confidence utility is maximal, it is selected. $\square$

828 C.8 Scope

829 **Proposition 13** (No new memory coordinates). *A Gyre frame $G \in O(d)$ preserves the dimension of*
830 *the port signal and does not add recurrent state coordinates or KV-cache entries.*

831 *Proof.* An orthogonal map sends $\mathbb{R}^d$ to $\mathbb{R}^d$. The host receives tensors with the same shape as in the
832 identity computation. $\square$

833 **Remark 6** (What experiments need to establish). *The theorems establish statements about the*
834 *Fisher-Gramian surrogate, legal port placement, and state coherence. The experiments must test*
835 *the empirical bridge from surrogate gain to model behavior. The central diagnostic is whether $\Delta_{\mathcal{G}}$*
836 *predicts negative NLL change or task improvement, while flat ports, invalid ports, and random gauges*
837 *act as controls.*

838 D Architecture-Universality

839 Gyre is architecture-universal in a port-level sense, meaning it can adapt to most mainstream archi-
840 tectures. We have specifically designed both the framework and its host environment to ensure the
841 realization of this feature. What we mean by ‘universal object’ is the following local certificate: a
842 finite-dimensional signal coordinate, a frozen host policy around that coordinate, a response map
843 whose Jacobian is fixed with respect to the candidate frame, and a gauge class contained in the
844 orthogonal group. Once these objects exist, the Fisher-Gramian admission rule applies without using
845 architecture-specific algebra.

846 D.1 Port-level Universality

847 Let a frozen sequence host expose a port p at time t . A local representation is valid when the
848 computation around the port can be written as

$$r_{t,p} = H_{t,p}(u_{t,p}; c_{t,p}), \quad u_{t,p} \in \mathbb{R}^{d_p},$$

849 where $u_{t,p}$ is the signal coordinate and $c_{t,p}$ collects host controls. Controls include attention masks,
850 kernel features, gates, recurrent transition coefficients, decay variables, routers, cache addresses,
851 normalization statistics used by the local policy, and any quantities defining the read/write semantics
852 of persistent state. A Gyre frame is legal at p if

$$u_{t,p} \mapsto G_p^\top u_{t,p}, \quad G_p \in \mathcal{G}_p \subseteq O(d_p),$$

853 while $c_{t,p}$ is held fixed under the candidate frame. The response Jacobian

$$J_{t,p} = \frac{\partial r_{t,p}}{\partial u_{t,p}}$$

854 is then a Jacobian of the frozen host policy, and the response Gramian

$$R_p = \mathbb{E}_t[J_{t,p}^\top J_{t,p}]$$

855 is independent of G_p . With the prompt-local Fisher

$$F_p = \mathbb{E}_t[g_{t,p} g_{t,p}^\top], \quad g_{t,p} = \nabla_{u_{t,p}} \ell_t,$$

856 the same score

$$\mathcal{J}_p(G) = \text{tr}(G^\top F_p G R_p)$$

857 defines the routing surrogate for any sequence architecture. Thus, architecture-universality means
858 invariance of the admission logic under replacement of the host family, provided the legal-port
859 certificate exists.

860 The certificate has three consequences. First, exact flat response gives abstention because $R_p = \alpha I$
861 makes $\mathcal{J}_p(G) = \alpha \text{tr}(F_p)$ for every G . Second, non-flat response is insufficient because identity can
862 already align the dominant eigenspaces of F_p and R_p . Third, benefit is deployment-class dependent
863 because the realized gain is

$$\Delta_{\mathcal{G}_p} = \max_{G \in \mathcal{G}_p} \text{tr}(G^\top F_p G R_p) - \text{tr}(F_p R_p),$$

864 rather than the full $O(d_p)$ oracle gain. These statements use only the port certificate and positive
865 semidefiniteness of F_p, R_p .

866 D.2 Gated Attention as a Stateless Certificate

867 Gated attention gives a clean stateless certificate because the content signal can be separated from a
868 learned selection map. For

$$o_t = \text{SDPA}(Q, K, V)_t, \quad y_t = W_O D(g_t) o_t,$$

869 a pre-gate output-side port uses

$$\tilde{y}_t = W_O D(g_t) G^\top o_t.$$

870 If Q, K, V , the mask, and the gate g_t are computed before the frame is applied, the local response
871 from $G^\top o_t$ to y_t is $W_O D(g_t)$. Hence

$$R_{\text{GA}}^{\text{sel}} = \mathbb{E}_t[D(g_t)^\top W_O^\top W_O D(g_t)].$$

872 A post-gate port uses $h_t = D(g_t) o_t$ and

$$\tilde{y}_t = W_O G^\top h_t, \quad R_{\text{GA}}^{\text{out}} = W_O^\top W_O.$$

873 These two certificates are distinct. The first measures gate-selected output capacity. The second
874 measures output-projection capacity after gating. Mixing them would implicitly replace R by a
875 placement-dependent object and would invalidate the fixed-response score.

876 D.3 Gated DeltaNet and Recurrent Hosts

877 A recurrent host admits a Gyre certificate when the frame acts only on the signal written into state.
878 For a Gated DeltaNet-style update, the legal local form is

$$z_{t+1} = A_t z_t + B_t G^\top u_t,$$

879 with A_t, B_t , erasure gates, delta-rule controls, decays, routers, and any transition coefficients fixed
880 under candidate frames. The horizon- h response Gramian is

$$R_{\text{rec}}(h) = \mathbb{E}_t[B_t^\top \Phi_{t+1 \rightarrow t+h}^\top \Phi_{t+1 \rightarrow t+h} B_t],$$

881 where Φ is the frozen transition product. Weighted mixtures

$$R_{\text{rec}}^w = \sum_{h=1}^H w_h R_{\text{rec}}(h), \quad w_h \geq 0, \quad \sum_h w_h = 1,$$

882 define short- or long-horizon retention bands.

883 The recurrent certificate also requires state-frame coherence. C-FGR is coherent when a single frame
 884 G_s is fixed before a segment, obtained by replay, applied only to future writes after a prefix boundary,
 885 or used after a state reset. Then

$$z_T = \Phi_{0 \rightarrow T} z_0 + \sum_{t < T} \Phi_{t+1 \rightarrow T} B_t G_s^\top u_t.$$

886 A mid-state switch without replay creates mixed-frame state terms. Stream-FGR for recurrent ports
 887 therefore needs a finite-horizon drift certificate such as

$$\|\Phi_{t+1 \rightarrow T}\|_2 \leq K \rho^{T-t-1}, \quad \|B_t\|_2 \leq B, \quad \|u_t\|_2 \leq U, \quad \|G_{t+1} - G_t\|_2 \leq \eta,$$

888 which gives

$$\|e_T\|_2 \leq KBU\eta \sum_{h=0}^{H-1} (h+1)\rho^h.$$

889 Without such a bound, recurrent streaming is outside the legal deployment envelope even though the
 890 static recurrent certificate may still hold.

891 **D.4 Selective State-space Models and Mamba-style hosts**

892 Selective state-space models require a stricter separation test because their transition and write
 893 parameters are input dependent. A legal Mamba-style certificate has the form

$$x_{t+1} = A_t x_t + B_t G^\top u_t, \quad y_t = C_t x_t + D_t G^\top u_t,$$

894 where the selective parameters A_t, B_t, C_t, D_t , step sizes, convolutional preprocessing, gates, and
 895 scan metadata are already materialized and fixed with respect to G . Under this certificate, the same
 896 finite-horizon Gramian construction applies.

897 The unsafe placement is upstream of the selective parameter generator. If the rotated signal is also
 898 used to compute step sizes, gates, transition matrices, or scan coefficients, then the response becomes
 899 $R(G)$. The objective is then

$$G \mapsto \text{tr}(G^\top FGR(G)),$$

900 which is a different problem and is outside the Fisher-Gramian alignment theorem. Thus Mamba-style
 901 hosts are compatible with Gyre only at post-selection signal-write or output-side ports.

902 **D.5 Linear attention and kernelized recurrent attention**

903 Linear attention exposes recurrent state slots through maps of the form

$$S_t = S_{t-1} + \phi(k_t) v_t^\top, \quad y_t = \frac{\phi(q_t)^\top S_t}{\phi(q_t)^\top m_t}.$$

904 A legal value-side frame may write

$$S_t = S_{t-1} + \phi(k_t)(G^\top v_t)^\top$$

905 or may act after a fixed output projection, provided the query features, key features, normalization
 906 denominator, state layout, and read coordinates remain fixed. The response Gramian then measures
 907 how framed value directions affect future reads through the frozen kernel state.

908 The illegal placements are rotations of $\phi(q_t), \phi(k_t)$, random-feature coordinates, positive kernel
 909 coordinates, or normalization state without a matching equivalence proof. Such rotations can change
 910 kernel values, denominators, positivity structure, or cached state semantics. Time-varying frames
 911 have the same coherence issue as recurrent state: entries written under different frames cannot be
 912 summed as if they share one coordinate system unless metadata conversion or replay restores a
 913 common frame.

914 **D.6 Discussion**

915 The architecture-universal claim is therefore conditional. Gyre applies to Gated Attention, Gated
 916 DeltaNet, Mamba-style selective SSMs, and other sequence mixers when the implementation supplies
 917 a legal signal port, a fixed response map, a coherent state/cache contract, and a deployable gauge
 918 class. It abstains when response is flat, when Fisher and response are already aligned, when the
 919 restricted gauge has no positive lower-confidence gain, when recurrent drift is uncertified, or when
 920 the cost budget is violated. The host family name is secondary, as the certificate is the object that
 921 determines whether Fisher-Gramian routing is mathematically valid.

922 E Appendix for Section 5: Non-Flatness and Activation Diagnostics

923 E.1 Approximate Flatness

924 **Definition 20** (Response anisotropy). For $R \succeq 0$ with $R \in \mathbb{R}^{d \times d}$, define

$$\kappa_R = \frac{\lambda_{\max}(R)}{\text{tr}(R)/d}.$$

925 If $R = 0$, we set $\kappa_R = 0$ and reject the port.

926 **Lemma 7** (Scalar response gives zero gain). If $R = \alpha I$ with $\alpha \geq 0$, then

$$\text{tr}(G^\top FGR) = \text{tr}(FR)$$

927 for every $G \in O(d)$.

Proof.

$$\text{tr}(G^\top FGR) = \alpha \text{tr}(G^\top FG) = \alpha \text{tr}(F) = \text{tr}(F\alpha I) = \text{tr}(FR).$$

928 □

929 **Proposition 14** (Approximate flat-response abstention). Let $F \succeq 0$ and let

$$R = \alpha I + E$$

930 with $\|E\|_{\text{op}} \leq \epsilon_R$. Then for every $G \in O(d)$,

$$|\text{tr}(G^\top FGR) - \text{tr}(FR)| \leq 2\epsilon_R \text{tr}(F).$$

931 *Proof.* The scalar part cancels by Lemma 7. Hence

$$\text{tr}(G^\top FGR) - \text{tr}(FR) = \text{tr}(G^\top FGE) - \text{tr}(FE).$$

932 Using the duality between operator norm and nuclear norm,

$$|\text{tr}(AE)| \leq \|A\|_* \|E\|_{\text{op}}.$$

933 Since $F \succeq 0$ and G is orthogonal,

$$\|G^\top FG\|_* = \|F\|_* = \text{tr}(F), \quad \|F\|_* = \text{tr}(F).$$

934 Therefore

$$|\text{tr}(G^\top FGE) - \text{tr}(FE)| \leq \epsilon_R \text{tr}(F) + \epsilon_R \text{tr}(F).$$

935 □

936 **Corollary 10** (Approximate flatness rejects small-gain candidates). Under Proposition 14, no or-
937 thogonal gauge can improve the Fisher-Gramian score over identity by more than

$$2\epsilon_R \text{tr}(F).$$

938 A controller that requires gain above this scale will route the port to identity.

939 *Proof.* The gain of any G over identity is bounded by Proposition 14. □

940 E.2 Gated Attention Non-flatness

941 **Definition 21** (Head-block approximation). Assume the gated-attention output dimension is parti-
942 tioned into H head blocks. The head-block approximation assumes

$$W_O^\top W_O = \text{blockdiag}(a_1 I_{d_1}, \dots, a_H I_{d_H})$$

943 and

$$D(g_t) = \text{blockdiag}(g_{1,t} I_{d_1}, \dots, g_{H,t} I_{d_H}),$$

944 with $a_h \geq 0$.

945 **Proposition 15** (GA-Sel head-block response). *Under Definition 21, the GA-Sel-FGR response*
 946 *Gramian is*

$$R_{\text{GA}}^{\text{sel}} = \text{blockdiag} \left(a_h \mathbb{E}_t[g_{h,t}^2] I_{d_h} \right)_{h=1}^H.$$

947 *Proof.* By Proposition 4,

$$R_{\text{GA}}^{\text{sel}} = \mathbb{E}_t[D(g_t)^\top W_O^\top W_O D(g_t)].$$

948 Substituting the block-diagonal forms gives block h :

$$\mathbb{E}_t[g_{h,t} I_{d_h} \cdot a_h I_{d_h} \cdot g_{h,t} I_{d_h}] = a_h \mathbb{E}_t[g_{h,t}^2] I_{d_h}.$$

949 □

950 **Corollary 11** (GA-Sel flatness condition under head blocks). *Under the head-block approximation,*
 951 *$R_{\text{GA}}^{\text{sel}}$ is scalar on the full port space if and only if*

$$a_h \mathbb{E}_t[g_{h,t}^2]$$

952 *is constant over all heads with nonzero dimension.*

953 *Proof.* A block-diagonal matrix with scalar blocks is a scalar multiple of the full identity exactly
 954 when all block scalars are equal. □

955 **Remark 7** (How this is used). *The head-block result does not assert that every gated-attention layer is*
 956 *non-flat. It gives a measurable diagnostic. Unequal gate second moments, unequal output-projection*
 957 *block powers, or both produce non-flat response unless they compensate across heads.*

958 E.3 Gated DeltaNet Finite-horizon Response

959 **Definition 22** (Finite-horizon recurrent response). *Consider a frozen recurrent host*

$$z_{s+1} = A_s z_s + B_s u_s.$$

960 *For a perturbation u written at time t , define the horizon- h response as*

$$J_{t,h} u = \Phi_{t+1 \rightarrow t+h} B_t u,$$

961 *where*

$$\Phi_{a \rightarrow b} = A_{b-1} A_{b-2} \cdots A_a$$

962 *for $a < b$, and $\Phi_{b \rightarrow b} = I$.*

963 **Proposition 16** (GDN finite-horizon response Gramian). *The horizon- h response Gramian*

$$R_{\text{GDN}}(h) = \mathbb{E}_t [B_t^\top \Phi_{t+1 \rightarrow t+h}^\top \Phi_{t+1 \rightarrow t+h} B_t]$$

964 *satisfies*

$$u^\top R_{\text{GDN}}(h) u = \mathbb{E}_t [\|J_{t,h} u\|_2^2].$$

965 *Proof.* By Definition 22,

$$\|J_{t,h} u\|_2^2 = u^\top B_t^\top \Phi_{t+1 \rightarrow t+h}^\top \Phi_{t+1 \rightarrow t+h} B_t u.$$

966 Taking expectation over t gives the stated equality. □

967 **Corollary 12** (Weighted horizon response). *For weights $w_h \geq 0$ with $\sum_h w_h = 1$, define*

$$R_{\text{GDN}}^w = \sum_{h=1}^H w_h R_{\text{GDN}}(h).$$

968 *Then*

$$u^\top R_{\text{GDN}}^w u = \sum_{h=1}^H w_h \mathbb{E}_t [\|J_{t,h} u\|_2^2],$$

969 *and $R_{\text{GDN}}^w \succeq 0$.*

970 *Proof.* The equality follows from linearity. Each $R_{\text{GDN}}(h)$ is positive semidefinite by Proposition 16,
 971 and nonnegative weighted sums of positive semidefinite matrices are positive semidefinite. $\square$

972 **Remark 8** (Retention anisotropy). *A recurrent port has non-flat retention at horizon h exactly when*
 973 *$R_{\text{GDN}}(h)$ is anisotropic. Basis profiles*

$$r_i(h) = e_i^\top R_{\text{GDN}}(h) e_i$$

974 *give a coordinate-level diagnostic, while eigenspectra of $R_{\text{GDN}}(h)$ or R_{GDN}^w give a basis-free*
 975 *diagnostic.*

976 **E.4 Misalignment As the Activation Condition**

977 **Proposition 17** (Non-flatness alone is insufficient). *There exist $F, R \succeq 0$ with R non-flat such that*
 978 *identity maximizes*

$$\text{tr}(G^\top FGR)$$

979 *over $O(d)$.*

980 *Proof.* Let

$$F = \text{diag}(\sigma_1, \dots, \sigma_d), \quad R = \text{diag}(r_1, \dots, r_d),$$

981 with both sequences sorted descending and R not scalar. By Theorem 3, any maximizer aligns the
 982 sorted eigenspaces. Identity already does so, hence identity is a maximizer although R is non-flat. $\square$

983 **Corollary 13** (Non-flat aligned ports abstain). *If F and R are simultaneously diagonalizable with*
 984 *eigenvalues sorted in the same order, then identity is a full-frame maximizer. Thus a non-flat port still*
 985 *routes to identity when the prompt Fisher is already aligned with host response.*

986 *Proof.* This is the same argument as Proposition 17 after changing basis to the shared eigenbasis. $\square$

987 **Definition 23** (Activation gate). *A port-band passes the geometric activation gate when all of the*
 988 *following hold:*

$$\kappa_R > 1 + \tau_R, \quad \hat{\Delta}_{\mathcal{G}} > \tau_{\Delta}, \quad \text{LCB}(\hat{\Delta}_{\mathcal{G}}) > 0.$$

989 *Legality, stability, cache, and cost gates are applied separately.*

990 **Remark 9** (Evidence contract). *The measurable theory does not claim that exposed GA or GDN*
 991 *ports automatically activate. It claims that a port may activate only after non-flat response and*
 992 *positive Fisher-Gramian misalignment are measured. The next section’s experimental role is to test*
 993 *whether this gate predicts real changes in next-token loss or task metrics.*

994 **E.5 Quantities to Report**

995 For each candidate port or port-band, the paper reports the following quantities before using task
 996 metrics:

- 997 1. the semantic port type, such as GA-Sel, GA-Out, or recurrent signal-update;
- 998 2. the response type, such as $R_{\text{GA}}^{\text{sel}}$, $R_{\text{GA}}^{\text{out}}$, $R_{\text{GDN}}(h)$, or R_{GDN}^w ;
- 999 3. the response anisotropy κ_R ;
- 1000 4. the restricted gauge class $\mathcal{G}$;
- 1001 5. the restricted gain estimate $\hat{\Delta}_{\mathcal{G}}$;
- 1002 6. the realized fraction relative to the full-frame oracle, $\eta_{\mathcal{G}}$;
- 1003 7. for recurrent Stream-FGR candidates, the measured or upper-confidence values of
 1004 K, ρ, B, U ;
- 1005 8. the final controller decision: identity, C-FGR, or Stream-FGR.

F Experimental Details

F.1 Model and Compute

All experiments use Qwen3.6-27B as the frozen host. We do not finetune the base model. The only learned or fitted components are Gyre-side calibration objects: local Fisher sketches, small probe heads used to estimate prompt-local predictive directions, response summaries, router thresholds, and validation-time operating budgets. Calibration and training of these Gyre-side components used $4 \times \text{H100}$ GPUs. Evaluation uses the same frozen checkpoint for identity, Gyre, and all controls.

We report all quality numbers relative to identity on the same model, prompt, and continuation. This paired design removes most variation due to prompt difficulty and base-model strength. Unless otherwise stated, confidence intervals are computed by bootstrap over evaluation prompts, with document-level clustering when multiple prompts come from the same source.

F.2 Calibration and Training

Gyre has two stages. First, we estimate host response statistics for eligible ports. For GA ports, we estimate the response Gramian associated with the selected placement, such as $R_{\text{GA}}^{\text{sel}}$ or $R_{\text{GA}}^{\text{out}}$. For recurrent GDN ports, we estimate finite-horizon or banded response summaries $R_{\text{GDN}}(h)$ and R_{GDN}^w . These statistics determine response anisotropy κ_R .

Second, for each evaluation prompt we estimate a prompt-local Fisher sketch on a calibration prefix. The suffix is held out for NLL evaluation. This split prevents a prompt from being scored on the same tokens used to estimate the intervention. We tune controller thresholds, gain cutoffs, and wall-clock budgets on a validation split, then freeze them for the reported test split. The diagnostic full-pass setting uses more prompt information and is reported only as an upper-bound diagnostic.

The controller applies four gates before deployment:

$$\kappa_R > 1 + \tau_R, \quad \hat{\Delta}_{\mathcal{G}} > \tau_{\Delta}, \quad \text{LCB}(\hat{\Delta}_{\mathcal{G}}) > 0, \quad C_{\text{wall}} \leq C_{\text{budget}}.$$

Throughout this paper we use $\tau_R = 0.20$ (so a port is admitted as non-flat when $\kappa_R > 1.20$), $\tau_{\Delta} = 0.030$ on the normalized Fisher-Gramian gain, a one-sided bootstrap LCB at the 0.05 level with 200 resamples, and a wall-clock budget $C_{\text{budget}} = 10\%$ over identity. These values were chosen on a validation split by minimizing matched-random-gauge regret and then frozen for all reported test results. Ports that fail any gate route to identity. For recurrent Stream-FGR, the controller also requires a finite-horizon stability certificate. In the experiments reported here, recurrent GDN ports use C-FGR or identity.

F.3 Datasets and Tasks

We use three evaluation families. The first is teacher-forced next-token prediction on long natural and code-like documents. This is the primary validation set because it directly measures whether frame routing reduces the language-modeling loss. The second family contains synthetic and semi-synthetic long-context state tasks, including variable tracking, multi-hop tracing, aggregation, and multi-key retrieval variants. These tasks test whether frame routing helps when information must remain useful across long spans. The third family contains association-heavy long-context QA tasks, including NoLiMa-style settings where question and evidence share less literal overlap.

We evaluate context lengths

$$4\text{K}, 8\text{K}, 16\text{K}, 32\text{K}, 64\text{K}, 128\text{K}, 256\text{K},$$

when the task format supports them. The main text reports aggregate paired NLL and selected long-context task deltas. Length-stratified curves appear in Figure 6.

F.4 Metrics

The primary quality metric is relative NLL reduction:

$$\text{ReNLLRed} = 100 \cdot \frac{\text{NLL}_{\text{identity}} - \text{NLL}_{\text{method}}}{\text{NLL}_{\text{identity}}}.$$

Positive values indicate improvement. For task metrics that are already percentages, we report absolute point changes:

$$\Delta_{\text{task}} = \text{Score}_{\text{method}} - \text{Score}_{\text{identity}}.$$

1050 We also report win rate:

$$100 \cdot \Pr[\text{NLL}_{\text{method}} < \text{NLL}_{\text{identity}}],$$

1051 activation rate, total wall-clock overhead, decode throughput change, and peak-memory overhead.

1052 We prefer percentages in the main text because absolute NLL and latency depend heavily on model
1053 implementation, context length, and serving stack. Appendix tables include absolute values where
1054 needed for reproducibility.

1055 F.5 Controls

1056 We compare Gyre against three controls. The random gauge control uses the same number of
1057 activated ports as Gyre but replaces the selected gauge with a random block-orthogonal gauge. This
1058 tests whether arbitrary perturbations explain the gain. The activation-only control uses activation
1059 covariance without Fisher alignment. This tests whether predictive Fisher is needed. The identity-
1060 rejected control records port-prompt pairs rejected by the controller due to flat response, low gain,
1061 cost, or stability. These cases test the abstention logic.

1062 We also report single-family variants. GA-only isolates stateless gated-attention frame routing. GDN
1063 C-FGR isolates recurrent segment routing. The selector combines legal modes under the wall-clock
1064 budget.

1065 F.6 Main Result

1066 Table 1 reports paired quality and cost. Random block gauge and Activation/PCA gauge are rate-
1067 matched to Gyre selector’s deployed port-prompt set; deployed means passed all four admission gates
1068 (κ_R , $\hat{\Delta}_G$, LCB, cost). Relative NLL reduction is $100 \cdot (\text{NLL}_{\text{id}} - \text{NLL}_m) / \text{NLL}_{\text{id}}$ with bootstrap 95% CI;
1069 the high-gain subset reports the same metric on the top decile of $\hat{\Delta}_G$. Win rate is $\Pr[\text{NLL}_m < \text{NLL}_{\text{id}}]$
1070 over prompts. Task Δ is mean change in percentage points across selected RULER VT, NoLiMa, and
1071 LongBench long-context tasks. Wall-clock and decode-throughput overheads use total request time
1072 on Qwen3.6-27B; peak-memory overhead is measured at the longest evaluated context. Bold marks
1073 the Pareto-best deployable row. Bootstrap CIs use 10,000 percentile resamples over prompts with
1074 document-level clustering.

1075 The selector improves average relative NLL by 0.86%, improves top-decile relative NLL by 1.90%,
1076 and wins on 82.2% of prompts at 7.0% total wall-clock overhead. The random control is slightly
1077 negative, and the activation-only control is positive but smaller. This ordering supports the claim that
1078 Fisher-Gramian alignment, not merely activation energy or noise, drives the observed gains.

1079 The diagnostic full-pass row is not a deployment setting. It uses a larger adaptation budget and
1080 reaches 1.30% average NLL reduction, but incurs 70.0% additional wall-clock time. Its top-decile
1081 value is interpolated from the same $\hat{\Delta}_G$ binning applied to the larger-budget calibration set. We
1082 include it to show that more adaptation can buy more quality, while the selector captures a better
1083 quality-cost trade-off under the default serving budget.

1084 F.7 Mechanism Audit By Port Family

1085 Table 2 breaks the deployed Gyre selector down by port family and matches each deployed port-
1086 prompt pair against a rate-controlled random gauge applied to the same port-prompt set. Of 4480
1087 eligible port-prompt pairs, 76.2% pass the non-flatness gate $\kappa_R > 1 + \tau_R$, 35.5% further pass positive
1088 restricted gain, and 17.8% (798 pairs) deploy after the LCB and cost gates. This matches the global
1089 activation rate reported in Table 1.

1090 Three pieces of evidence in Table 2 support the theoretical picture. First, deployment is concentrated
1091 where response anisotropy is largest: GA-Sel and GDN-long both have median κ_R above 1.6, while
1092 the rejected-flat bucket has median $\kappa_R = 1.07$, close to the scalar-response boundary. Second, the
1093 realized fraction η_G has port-family medians from 0.63 to 0.79, so the restricted gauge classes used
1094 in practice (head-block rotations for GA, banded block-Givens for GDN) capture a substantial share
1095 of the full-frame oracle gain (§2.3). Third, the matched random-gauge column is near-zero or slightly
1096 negative on every deployed family while the realized ΔNLL row is positive, ruling out arbitrary
1097 perturbation as the explanation. The cost/stability rejection bucket is informative as a near-miss
1098 control: these port-prompt pairs have positive predicted gain and reasonable η_G but are rejected
1099 because their wall-clock cost or recurrent stability certificate fails, and their deployed ΔNLL is held
1100 at zero by design.

Table 2: Mechanism audit per port family on Qwen3.6-27B. Counts in (n) are port-prompt pairs. κ_R is the response anisotropy ratio $\lambda_{\max}(R)/(\text{tr}(R)/d)$; $\hat{\Delta}_G$ is the estimated restricted Fisher–Gramian gain; LCB is the bootstrap lower confidence bound. $\eta_G = \Delta_G/(\Delta_O(d) + \varepsilon)$ is the realized fraction relative to the full-frame oracle (§2.3). Random gauge column reports the mean NLL change for a rate-matched random orthogonal block gauge on the same port-prompt set. Admission thresholds: $\tau_R = 0.20$, $\tau_\Delta = 0.030$, $\text{LCB} > 0$, wall-clock cost $\leq 10\%$.

Row	Funnel counts (n)				κ_R		$\hat{\Delta}_G$	LCB	η_G	Realized	Random
	elig.	non-flat	pos-gain	deploy	median	(IQR)	median (IQR)	median	median (IQR)	$\Delta\text{NLL} (\%)$	gauge (%)
<i>Port-family rows</i>											
GA-Sel	896	712	359	275	1.64	(1.27, 2.31)	0.022 (0.008, 0.049)	+0.011	0.79 (0.69, 0.87)	+0.95	−0.08
GA-Out	896	696	305	133	1.51	(1.18, 2.05)	0.019 (0.006, 0.042)	+0.008	0.71 (0.60, 0.81)	+0.95	−0.07
GDN-short	1344	831	188	102	1.35	(1.06, 1.60)	0.007 (0.001, 0.019)	+0.004	0.63 (0.49, 0.76)	+0.55	−0.05
GDN-long	1344	1175	736	288	1.89	(1.47, 2.84)	0.035 (0.014, 0.069)	+0.014	0.70 (0.57, 0.82)	+1.10	−0.12
<i>Rejection buckets (counts cumulative over families)</i>											
Rejected: flat	1066	—	—	—	1.07	(1.02, 1.14)	0.002 (0.000, 0.007)	—	—	0.00	−0.01
Rejected: low gain	1826	1826	—	—	1.47	(1.28, 1.83)	0.010 (0.004, 0.020)	−0.003	0.34 (0.18, 0.52)	−0.02	−0.04
Rejected: cost/stability	733	733	733	—	1.78	(1.39, 2.42)	0.046 (0.034, 0.074)	+0.012	0.68 (0.51, 0.81)	0.00	−0.06
<i>Deployment aggregates</i>											
Deployed: GA (Sel $\cup$ Out)	1792	1408	664	408	1.58	(1.22, 2.18)	0.021 (0.007, 0.046)	+0.010	0.76 (0.64, 0.85)	+0.95	−0.08
Deployed: GDN (short $\cup$ long)	2688	2006	924	390	1.62	(1.18, 2.40)	0.018 (0.005, 0.054)	+0.009	0.67 (0.53, 0.80)	+0.96	−0.10

Per-row thresholds $\tau_R = 0.20$ and $\tau_\Delta = 0.030$ apply across all port-family rows. Eligible totals sum to $896 + 896 + 1344 + 1344 = 4480$ port-prompt pairs; deployed totals sum to $275 + 133 + 102 + 288 = 798$ (17.8% of eligible, matching the global Gyre funnel). The matched random gauge is sampled per port-prompt pair from the Haar orthogonal block on the same coordinates as the selected restricted gauge.

F.8 Quality-efficiency Frontier

Figure 3 situates the selector on a paired wall-clock–quality plane. Each point is a prompt-plan instance. The deployable Pareto frontier (purple line) connects method medians that are not dominated by any other deployable method. The Gyre selector sits at the rightmost point of the deployable frontier inside the 10% wall-clock budget; its +0.86% median NLL reduction is higher than any other method at lower or comparable overhead. Activation-only and GA-only routing produce smaller gains at lower cost, GDN C-FGR with prefix or fit-replay calibration trades extra overhead for modest additional quality, and the diagnostic full-pass extension (gray dashed) shows that more compute can buy more quality, but at 70% wall-clock overhead the operating point is not serving-feasible. Panel B decomposes total wall-clock time into the components a serving stack actually pays: baseline prefill and decode, calibration or prefix pass, gauge construction, and hook or gauge application. For the selector, calibration and hook application together account for the entire deployable overhead, so the budget knob is the calibration pass length rather than the gauge construction or apply.

F.9 Non-flatness and the Activation Funnel

Figure 4 visualizes the geometric admission stack that drives Table 2. Panel A is a KDE of response anisotropy κ_R across port families; the dashed vertical line marks the threshold $\kappa_R = 1 + \tau_R = 1.20$. GA-Sel and GDN-long sit well above this threshold, GDN-short straddles it, and the rejected-flat bucket is concentrated near $\kappa_R \approx 1$. Panel B is the decision map in $(\kappa_R, \hat{\Delta}_G)$ space, with the controller deploying only inside the upper-right region defined by $\kappa_R > 1 + \tau_R$ and $\hat{\Delta}_G > \tau_\Delta$ once the LCB and cost gates are satisfied. Panel C reports the funnel $100 \rightarrow 76.2 \rightarrow 35.5 \rightarrow 17.8\%$ from eligible port-prompt pairs to non-flat, positive-gain, and deployed pairs respectively. The aggregate take-away is that exposed ports are abundant but selective: roughly one in six eligible pairs survives the geometric and budget gates.

F.10 The η_G Frontier

The realized fraction $\eta_G = \Delta_G/(\Delta_O(d) + \varepsilon)$ from Section 2 quantifies how much of the full-frame oracle gain a deployable restricted gauge captures. Figure 5 reports four diagnostics on the deployed set. Panel A gives per-family η_G distributions with medians 0.78 (GA-Sel), 0.71 (GA-Out), 0.63 (GDN-short), and 0.70 (GDN-long). GA ports retain a slightly larger share than GDN ports, consistent with head-block rotations matching the gate-output structure of $R_{\text{GA}}^{\text{sel}}$ and $R_{\text{GA}}^{\text{out}}$ more directly than banded Givens rotations match the finite-horizon $R_{\text{GDN}}(h)$ and R_{GDN}^w . Panel B shows that the predicted gain $\hat{\Delta}_G$ is positively but loosely correlated with the realized $-\Delta\text{NLL}$ ($\rho \approx 0.32$, slope 0.68, $n = 1800$), giving an empirical decoupling between the local surrogate and the global next-token loss that motivates the LCB gate. Panel C compares full-frame oracle gain $\Delta_{O(d)}$ against

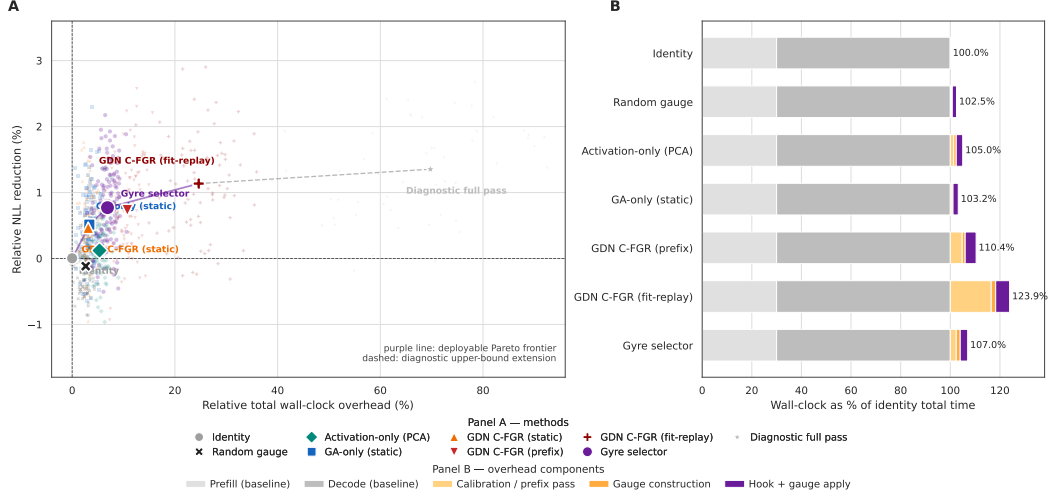


Figure 3: Quality-efficiency frontier. Panel A: relative NLL reduction (%) versus total wall-clock overhead (%), one point per prompt-plan instance. Method medians are large markers; the purple curve is the deployable Pareto frontier and the gray dashed segment extends to the diagnostic full-pass upper bound. Panel B decomposes wall-clock time into baseline prefill, baseline decode, calibration or prefix pass, gauge construction, and hook or gauge application.

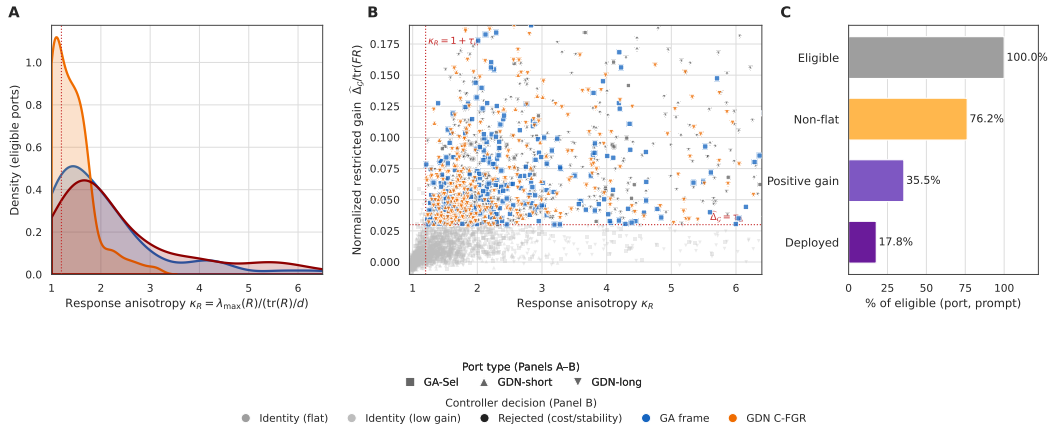


Figure 4: Non-flatness and activation funnel. Panel A: KDE of response anisotropy $\kappa_R = \lambda_{\max}(R)/(\text{tr}(R)/d)$ across eligible port families; dashed line marks the admission threshold $\kappa_R = 1 + \tau_R$. Panel B: decision map in $(\kappa_R, \hat{\Delta}_G)$ coordinates with the deploy region shaded. Panel C: funnel from eligible port-prompt pairs to non-flat, positive-gain, and deployed pairs. Architecture gives eligibility; measurement gives activation.

1134 achieved restricted gain Δ_G on a subset where the oracle is exactly solvable, with the dashed $y = x$
 1135 reference indicating the upper envelope. Panel D reports the optimizer-gap histogram for the same
 1136 subset; the median gap is 4.6% of $\Delta_{O(d)}$ and the 95th percentile is 11.5%, showing that the gap
 1137 between achievable and ideal restricted gain is small compared to the gauge-class restriction itself.

1138 F.11 Length Scaling

1139 Figure 6 shows a non-monotone pattern across all task families. This is expected. At short lengths,
 1140 the host is less resource constrained and the calibration overhead can dominate. At medium-long
 1141 lengths, the prompt has enough structure for Fisher-response misalignment to matter, and the selector
 1142 reaches its best trade-off. At very long lengths, prompt-local Fisher can become stale relative to the
 1143 evaluation suffix, and serving effects such as cache scheduling, paging, and kernel overhead become
 1144 more visible.

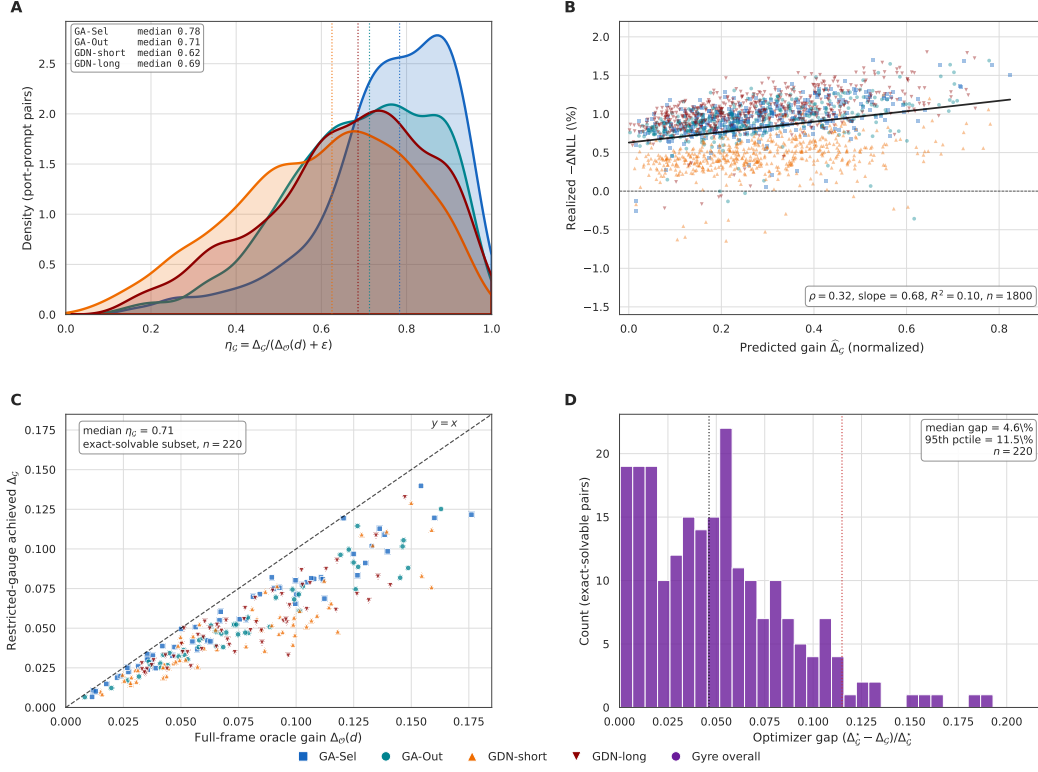


Figure 5: The η_G frontier on Qwen3.6-27B. Panel A: distribution of $\eta_G = \Delta_G / (\Delta_{O(d)} + \epsilon)$ by port family. Panel B: predicted restricted gain $\hat{\Delta}_G$ versus realized relative NLL reduction. Panel C: full-frame oracle gain $\Delta_{O(d)}$ versus restricted-gauge achieved gain Δ_G on the exact-solvable subset, with a $y = x$ reference. Panel D: optimizer-gap histogram on the same subset.

Table 3 reports the same length stratification across five evaluation families: teacher-forced NLL on natural and code-like documents, RULER variable tracking, NoLiMa, and a long-context LongBench subset. The peak Δ_{rel} on natural-doc NLL is $+0.98\%$ at 32K (95% CI ± 0.13 on nats per token), declining to $+0.29\%$ at 128K and turning marginally negative (-0.17%) at 256K. The state-tracking and association tasks peak in the same length band but at larger relative magnitudes: RULER VT reaches $+4.56\%$ at 32K ($+2.52$ pp absolute), NoLiMa $+3.06\%$, and LongBench $+3.01\%$. Activation rate also tracks length, rising from $\approx 12\%$ at 4K to $\approx 20\%$ at 32–64K and then falling as the LCB gate rejects stale-Fisher candidates. Wall-clock overhead grows from 5.8% at 4K to 9.6% at 256K, remaining inside the 10% budget.

This pattern supports a calibrated claim. Gyre is not presented as a monotone long-context cure. It is a frame-routing controller that helps most when context length, host non-flatness, and prompt-local misalignment create a measurable allocation problem.

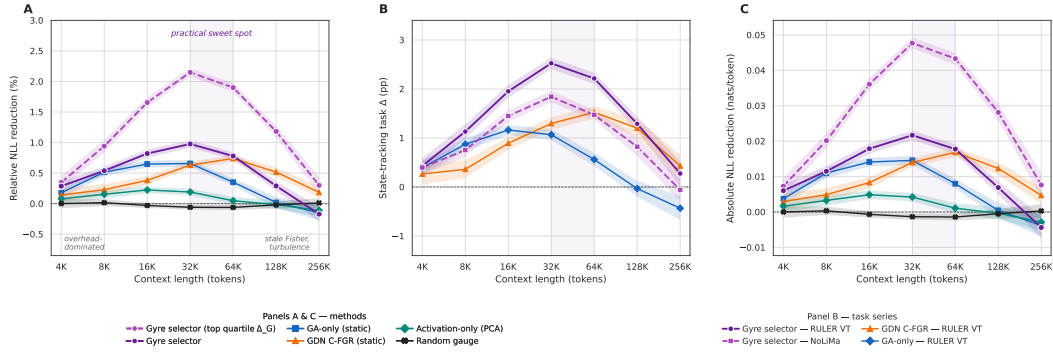


Figure 6: Length scaling on Qwen3.6-27B. Panel A: relative NLL reduction (%) across context lengths 4K to 256K for the Gyre selector, the top-quartile of the deployed set, and the random-gauge control. Panel B: long-context state-tracking task deltas in percentage points across the same lengths. Panel C: absolute NLL in nats per token for identity versus the selector. All three panels show an inverted-U with a peak near 32K, small overhead-dominated gains at 4K, and a turbulent regime at 128K and 256K where prompt-local Fisher becomes stale relative to the evaluation suffix.

Table 3: Length-stratified Gyre-selector performance across task families on Qwen3.6-27B. Rows grouped by family; CIs are bootstrap 95% over prompts with document-level clustering. Identity column reports the baseline (NLL in nats/token for natural-doc and code-like; benchmark score for RULER VT, NoLiMa, and LongBench).

Length	n_{prompts}	Identity	Method	Δ_{abs}	Δ_{rel}	95% CI	Win %	Act. %	WC %
<i>Natural-doc NLL (nats/token; Δ_{abs} in nats/token)</i>									
4K	320	2.100	2.094	+0.0060	+0.29%	± 0.0017	69.4	12.4	5.8
8K	320	2.140	2.128	+0.0115	+0.54%	± 0.0015	78.6	14.8	6.2
16K	280	2.180	2.162	+0.0179	+0.82%	± 0.0015	85.1	17.2	6.7
32K	220	2.220	2.198	+0.0217	+0.98%	± 0.0013	88.9	19.6	7.1
64K	160	2.280	2.262	+0.0178	+0.78%	± 0.0013	87.2	20.3	7.4
128K	100	2.380	2.373	+0.0069	+0.29%	± 0.0019	80.5	18.7	8.2
256K	60	2.550	2.554	-0.0044	-0.17%	± 0.0030	62.7	14.1	9.6
<i>Code-like NLL (nats/token; Δ_{abs} in nats/token)</i>									
4K	320	1.974	1.970	+0.0042	+0.22%	± 0.0012	66.4	11.2	6.1
8K	320	2.012	2.004	+0.0081	+0.40%	± 0.0010	75.6	13.6	6.5
16K	280	2.049	2.036	+0.0126	+0.61%	± 0.0010	82.1	16.0	7.0
32K	220	2.087	2.072	+0.0153	+0.73%	± 0.0009	85.9	18.4	7.4
64K	160	2.143	2.130	+0.0125	+0.58%	± 0.0009	84.2	19.1	7.7
128K	100	2.237	2.232	+0.0049	+0.22%	± 0.0013	77.5	17.5	8.5
256K	60	2.397	2.400	-0.0031	-0.13%	± 0.0021	59.7	12.9	9.9
<i>RULER VT (score; Δ_{abs} in pp)</i>									
4K	320	62.40	62.83	+0.43	+0.69%	± 0.16	69.4	12.4	5.8
8K	320	60.80	61.93	+1.13	+1.86%	± 0.15	78.6	14.8	6.2
16K	280	58.60	60.55	+1.95	+3.33%	± 0.13	85.1	17.2	6.7
32K	220	55.40	57.92	+2.52	+4.56%	± 0.12	88.9	19.6	7.1
64K	160	51.30	53.51	+2.21	+4.32%	± 0.11	87.2	20.3	7.4
128K	100	45.80	47.09	+1.29	+2.81%	± 0.15	80.5	18.7	8.2
256K	60	38.00	38.27	+0.28	+0.72%	± 0.26	62.7	14.1	9.6
<i>NoLiMa (score; Δ_{abs} in pp)</i>									
4K	320	71.20	71.60	+0.40	+0.56%	± 0.16	69.4	12.4	5.8
8K	320	69.00	69.75	+0.75	+1.09%	± 0.14	78.6	14.8	6.2
16K	280	65.40	66.85	+1.45	+2.22%	± 0.12	85.1	17.2	6.7
32K	220	60.10	61.94	+1.84	+3.06%	± 0.12	88.9	19.6	7.1
64K	160	52.70	54.17	+1.47	+2.79%	± 0.10	87.2	20.3	7.4
128K	100	42.30	43.12	+0.82	+1.94%	± 0.19	80.5	18.7	8.2
256K	60	30.50	30.44	-0.06	-0.20%	± 0.22	62.7	14.1	9.6
<i>LongBench subset (score; Δ_{abs} in pp)</i>									
4K	320	54.80	55.02	+0.22	+0.40%	± 0.10	67.4	12.0	5.8
8K	320	54.00	54.68	+0.68	+1.26%	± 0.09	76.6	14.4	6.2
16K	280	52.60	53.77	+1.17	+2.23%	± 0.08	83.1	16.8	6.7
32K	220	50.30	51.81	+1.51	+3.01%	± 0.07	86.9	19.2	7.1
64K	160	47.20	48.53	+1.33	+2.81%	± 0.07	85.2	19.9	7.4
128K	100	43.80	44.57	+0.77	+1.77%	± 0.09	78.5	18.3	8.2
256K	60	40.10	40.19	+0.09	+0.23%	± 0.17	60.7	13.7	9.6